Tensor Ring Decomposition-Based Generalized and Efficient Nonconvex Approach for Hyperspectral Anomaly Detection

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***Abstract*— Anomaly detection in hyperspectral images (HSIs) aims to identify sparse, interesting anomalies against the background, which has become a significant topic in remote sensing. Although the existing tensor-based methods have achieved commendable performance to some extent, there is still room for further improvement. In combination with three key techniques, i.e., gradient map-based modeling, circular tensor ring (TR) unfolding, and nonconvex regularization, this article proposes a novel generalized nonconvex method for hyperspectral anomaly detection (HAD) tasks within the TR framework. For the implementation of our proposed approach, abbreviated as TR-GNHAD, we first develop an effective and reliable HAD model in virtue of two newly unified nonconvex regularizers. The first regularizer is devised under a new prior characterization paradigm, which has a strong ability to encode two insightful prior information underlying the HSI's background simultaneously, i.e., global low rankness and local smoothness. The other regularizer can well capture the structured sparsity of the abnormal component. Then, we derive an efficient optimization algorithm to solve the proposed model based on the alternating direction method of multipliers (ADMMs) framework. Experiments conducted on 12 HSI datasets illustrate that the proposed approach achieves highly competitive performance in both qualitative and quantitative metrics compared with several state-of-the-art HAD methods.**

***Index Terms*— Alternating direction method of multiplier (ADMM) algorithm, generalized nonconvex regularizers, global low rankness, group sparsity, hyperspectral anomaly detection (HAD), local smoothness, tensor ring (TR) decomposition.**

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I. Introduction

A HYPERSPECTRAL image (HSI) is a 3-D data cube that comprises 2-D spatial information used to depict the distribution of ground objects and 1-D spectral information that reveals the intrinsic properties of materials, and has been successfully applied in various fields, including restoration [1], fusion [2], super resolution [3], [4], [5], classification [6], [7], [8], [9], denoising [10], [11], [12], [13], target detection [14], [15], [16], and anomaly detection [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50], [51], [52]. Among them, hyperspectral anomaly detection (HAD), which aims to identify and separate anomalous objects from the background, has attracted increasing attention due to its significant role in military and civilian fields (e.g., camouflage recognition, precision agriculture, and military surveillance [53], [54]). In recent decades, scholars have invested significant resources into HAD, leading to the development of various effective HAD methods. The existing HAD algorithms can be classified into four categories: statistic-based methods [17], [18], [19], [20], [21], [22], deep learning-based methods [23], [24], [25], [26], [27], [28], [29], representation-based methods [30], [31], [32], [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], and decomposition-based methods [44], [45], [46], [47], [48], [49], [50], [51], [52].

A classical statistics-based HAD method known as Reed–Xiaoli (RX) was proposed in [17], which postulates the background pixels that adhere to a multivariate Gaussian distribution while anomaly pixels do not. The mean and covariance of the background distribution are first computed by using all the image pixels. Subsequently, the Mahalanobis distance-based approach is employed to measure the anomaly of the test pixels. An enhanced, locally adaptive RX detector was introduced in [18], which employs a dual sliding window and confines the calculation of the mean and covariance to the pixels within this window. Based on the detection principle of RX, some other improved RX-based methods have been proposed, such as kernel-RX (KRX) [19], cluster KRX [20], weighted-RX [21], and segmented-RX [22].

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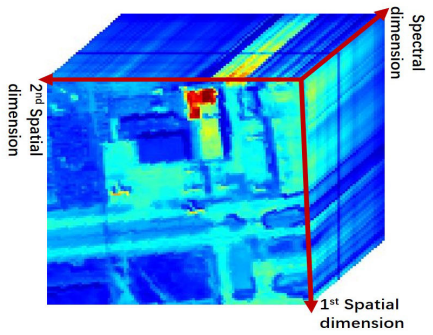


Fig. 1. Tensor format of HSI data that includes one spectral dimension and two spatial dimensions.

In addition, several collaborative representation and sparse representation methods were put forward for HAD in [30], [31], [32], and [33], found on the principle that the background pixels can be approximately represented by their neighborhood pixels, while the anomaly pixels cannot. With the development of deep learning technology, some methods grounded in this technique have also been introduced for HAD tasks, such as LREN [23], DeCNN-AD [24], PDBSNet [25], BS^{3}LNet [26], and LRR-Net [27].

In recent years, the utilization of low-rank and sparse matrix decomposition (LSMD) techniques has garnered wide attention in the HAD realm. One of the most popular LSMD-based methods is the robust principal component analysis (RPCA) [44], which decomposes an HSI into a low-rank matrix and a sparse anomaly matrix [45]. However, the RPCA fails to account for the influence of noise, which may result in a higher *false alarm rate*. To mitigate the impact of noise, a refined method called LSMAD was proposed in [46]. This method incorporated an additional noise matrix into the decomposition process, and then the Mahalanobis distance was utilized to precisely detect potential anomalies. Furthermore, under the assumption that the noise follows a mixture of Gaussian distributions, a new method called LMSD-MOG was proposed in [47]. Given the presence of numerous mixed pixels in the background of HSI data, the assumption that the background can be described by a single subspace may be inaccurate. A more reasonable solution is to consider the background as being located near the union of several subspaces. To this end, through the dictionary construction strategy, a low-rank and sparse representation (LRASA)-based method was designed in [34], which integrates the global and local structures of the background. To preserve the local geometrical structure and the spatial correlations of the background, a graph and total variation (TV) regularized low-rank representation (GTVLRR) approach was proposed in [35]. Recently, to obtain a more accurate approximation of the background and anomaly components, a unified nonconvex framework named HADGSM was developed in [36].

As shown in Fig. 1, HSI data can essentially be regarded as a three-order tensor, encompassing one spectral dimension and two spatial dimensions. Regrettably, the existing matrix-based detection methods often overlook the intrinsic multidimensional structure of HSI data, which limits their effectiveness in HAD tasks. To alleviate this problem, some HAD methods based on tensor representation [37], [38], [39], [40], [41], [42], [43] and tensor decomposition [48], [49], [50], [51], [52] have been proposed. These researches mostly focused on extracting the intrinsic prior structures (e.g., low-rankness, smoothness, and sparsity) of HSIs through different regularization methods. Among all prior structures, the low-rankness and smoothness emerge as the two most frequently utilized properties. For example, a tensor-based method called PTA [48] harnessed the spectral low-rank property and spatial smoothness of the background to improve the detection performance. To maximize the exploitation of the global low-rankness, a low average rank with a TV regularization model for HAD (LARTVAD) was proposed in [38]. Recently, a novel HAD method called TRDFTVAD was developed in [51], which adequately considers the low-rankness and smoothness properties of the background in all dimensions. Although the tensor-based approaches mentioned above can achieve better HAD performance over matrix-based methods, there still exists some room for further improvement. Specifically, the following observations were made.

1) Some studies utilized the typical convex surrogate, e.g., widely used tubal nuclear norm (TNN) [55], [56], [57] plus $\ell_{F,1}$-norm, to formulate the HAD model, which results in loose approximation and suboptimal solver for the original problem [38], [40].
2) Some nonconvex methods were proposed to further boost the HAD performance, yet they solely employed nonconvex approximation on low-rank terms [37], [39], [41], [42], [52]. Besides, they were solved case-by-case and lacked a universal form.
3) Some nonconvex approaches overlook other critical priors in HSI data beyond the global low-rankness [37], [39], [42], e.g., the local smoothness.
4) Many HAD investigations' performance, based on the low-rankness and smoothness priors, was greatly influenced by the trade-off parameters imposed between regularizers [38], [41], [43], [48], [51]. However, devising a universal guideline for precisely adjusting the balancing parameter remains challenging [50].
5) The majority of research has primarily focused on the Tucker factorization and tensor singular value decomposition (T-SVD) [37], [38], [39], [40], [41], [42], [43], [48], [49], [50], [52], with less attention given to the recently developed tensor ring (TR) framework [58].

To alleviate the aforementioned issues, this article first introduces a novel paradigm for the joint characterization of global low-rankness ($\mathbf{L}$) and local smoothness ($\mathbf{S}$) properties. This is primarily driven by several previous efforts that utilize gradient map modeling as a foundation [50], [59], [60], [61], [62], [63]. Specifically, by applying the difference operator to the original background tensor, we can convert its local smoothness to the sparsity of the resulting gradient maps. From the top-right side of Fig. 2, we observe that the proportion of elements close to zero in each gradient map is notably high, indicating that all gradient maps exhibit sparse configurations. Since the difference operator is essentially linear and does not alter the low-rank structure of the original

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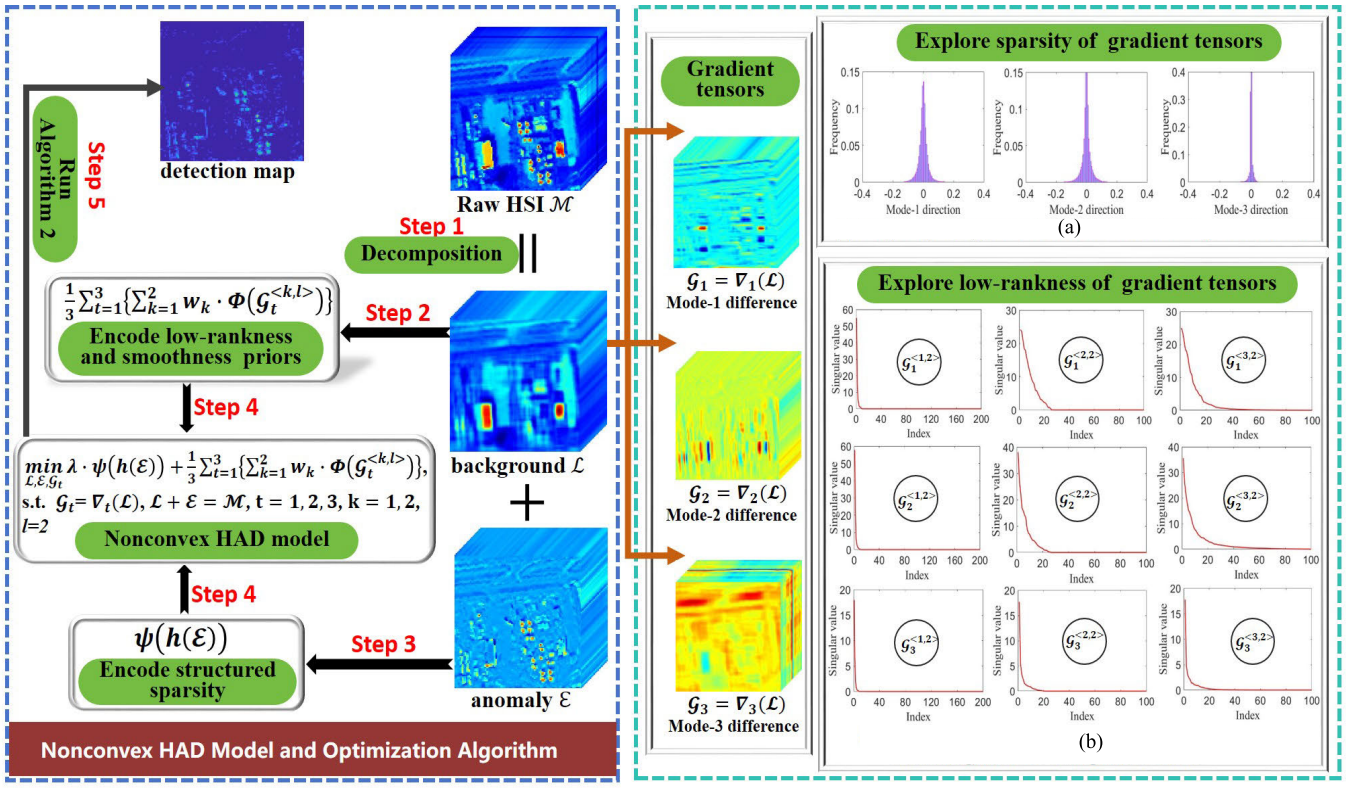


Fig. 2. Flowchart of our generalized nonconvex HAD method. *Right side:* a new prior characterization paradigm that equivalently transfers the **L** and **S** priors of the background to the low-rankness and sparsity of its gradient tensors. *Left side: Step 1:* Tensor RPCA: perform a joint low-rank and smooth plus sparse tensor decomposition from raw HSI. *Step 2:* Devise a unified nonconvex regularizer to encode the low-rankness and smoothness of the background. *Step 3:* Devise another nonconvex regularizer to encode the structured sparsity of the anomaly. *Step 4:* Propose a novel HAD model based on two nonconvex regularizers. *Step 5:* Solve the formulated model via Algorithm 2 derived from ADMM framework.

tensors, the resulting gradient maps inherently retain their low-rank properties. As shown in the bottom-right side of Fig. 2, all TR unfolding matrices of each gradient tensor exhibit strong low-rank property, which implies that all three gradient tensors equip with a low-ring-rank structure according to Theorem 3. Therefore, the difference operation transfers the **L** and **S** priors of the original background tensor to the low-rankness and sparsity of its gradient maps. Based on the above prior characterization scheme and the idea of tensor RPCA, we further investigate a novel generalized nonconvex HAD model within the TR framework. Finally, to obtain the accurate output of the anomaly detection maps, the well-known alternating direction method of multipliers (ADMM) framework [64] is adopted to optimize the formulated HAD model. By using circular TR unfolding and nonconvex constraints on the gradient maps of the background tensor, our novel approach is able to synchronously extract both the **L** and **S** priors with a single, concise, and unified regularizer. This is in contrast to the existing methods, which require multiple regularization terms to encode the **L** + **S** properties of HSI's background, and suffer from the difficulty of adjusting the trade-off parameters. In addition, instead of the basic convex forms such as the ℓ_1-norm and $\ell_{F,1}$-norm, another unified nonconvex regularization item is devised for characterizing the structured sparsity of the anomaly tensor. The practicability and effectiveness of the proposed HAD method are verified by extensive experiments.

A. Why Consider Utilizing TR Framework?

The advantages of using the TR framework for HAD tasks can be summarized as follows.

1) *Circular Dimension Permutation Invariance:* The TR decomposition approximates a high-dimensional tensor as a sequence of cyclically contracted low-dimensional core tensors organized in a ring structure. This ensures that the representation is invariant to the permutation of dimensions, which is crucial for HSI data where the order of spatial–spectral dimensions can vary. However, other decomposition frameworks, such as T-SVD, Tucker, and tensor train (TT), are highly dependent on the permutation of tensor dimensions.
2) *Enhanced Representation Ability:* The TR framework can be regarded as a generalized form of other tensor decompositions (e.g., TT decomposition), offering stronger representation capability. This enhanced ability allows it to better capture the complex structural characteristics in the background of HSI data, resulting in improved detection accuracy.
3) *Reduced Computational and Storage Complexity:* Compared with manipulating the whole original tensor data directly, performing relevant modeling and computations on the small-scale core factors derived from the TR framework can reduce the complexity of computation and storage. This is especially important for processing

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large-scale HSI datasets, which are usually characterized by extremely high dimensionality and large data volume.

B. Our Contributions

The main contributions of this article can be summarized as follows.

1) A novel generalized nonconvex method is developed for HAD tasks within the TR framework, which jointly integrates three pivotal technologies, i.e., gradient map-based modeling, circular TR unfolding, and nonconvex regularization. There are few related studies that have combined these ideas to accomplish HAD tasks.
2) Based on a new prior characterization paradigm, we build a unified nonconvex regularizer to encode the $\mathbf{L}+\mathbf{S}$ priors of background tensor synchronously. Compared with the existing HAD studies adopting the $\mathbf{L}$ and $\mathbf{S}$ priors separately, our strategy considers a more generalized, concise, and effective form. Furthermore, another unified regularizer involving a class of nonconvex functions is designed to capture the structured sparsity that existed in the anomaly tensor. Existing researches only consider the basic convex forms, such as ℓ_1-norm and $\ell_{F,1}$-norm. Finally, a novel generalized double nonconvex model is proposed for the HAD task.
3) In terms of algorithm design, we initially provide the corresponding methods for solving two critical subproblems. Assisted by the well-known ADMM framework, we then derive an efficient algorithm to handle the formulated HAD model. Experiments conducted on simulated and real HSI datasets show that the proposed method has a better detection performance in comparison to other classic HAD approaches.

C. Organization

The rest of this work is organized as follows. Some notations and preliminaries utilized in this article are presented in Section II. Section III introduces a generalized nonconvex framework includingthe model and algorithm for HAD tasks. Evaluation of experimental results and discussions constitute Section IV. Finally, we summarize our work in Section V.

II. Notations and Preliminaries

This section mainly summarizes the extensively used algebraic foundations under the TR framework. Please refer to the literatures [58], [65], [66], [67] for more details.

We first briefly review the definition of TR decomposition.

Definition 1 (TR Decomposition [58]): For an order-d tensor $\mathcal{A}\in\mathbb{R}^{n_1\times\cdots\times n_d}$ with TR rank $[r_1,\ldots,r_d]$, its TR decomposition is represented as a sequence of circularly contracted core tensor $\mathcal{U}^{(k)}\in\mathbb{R}^{r_k\times n_k\times r_{k+1}}$, $k=1,2,\ldots,d$, with $r_{d+1}=r_1$. Specifically, the elementwise relation of tensor $\mathcal{A}$ and its TR core tensors $\{\mathcal{U}^{(k)}\}_{k=1}^{d}$ is defined as follows:

$$\mathcal{A}_{i_1,\ldots,i_N}=\mathrm{Tr}\left(\prod_{k=1}^{d}\mathcal{U}^{(k)}(:,i_k,:)\right)$$

where $\mathcal{U}^{(k)}(:,i_k,:)\in\mathbb{R}^{r_k\times r_{k+1}}$ is the i_kth slice matrix of $\mathcal{U}^{(k)}$ along mode-2, and $\mathrm{Tr}(\cdot)$ is the matrix trace operator.

Based on the above definition, Huang et al. [66] and Yu et al. [67] proposed a new circular TR unfolding strategy.

Definition 2 (Circular TR Unfolding [66], [67]): Given an order-d tensor $\mathcal{A}\in\mathbb{R}^{n_1\times\cdots\times n_d}$, its TR unfolding (also called k-shifting l-unfolding) $\mathcal{A}^{\langle k,l\rangle}\in\mathbb{R}^{\prod_{i=k}^{k+l-1}n_i\times\prod_{j=k+l}^{k-1}n_j}$, $k,l\in\{1,2,\ldots,d\}$, is a matrix whose elements are defined via the relationship $(\mathcal{A}^{\langle k,l\rangle})_{s,t}=\mathcal{A}_{i_1,\ldots,i_d}$ with

$$s=1+\sum_{c=k}^{k+l-1}(i_c-1)\prod_{j=k}^{c-1}n_j,\quad t=1+\sum_{c=k+l}^{k-1}(i_c-1)\prod_{j=k+l}^{c-1}n_j.$$

In practice, $\mathcal{A}^{\langle k,l\rangle}$ can be constructed by first permuting $\mathcal{A}$ with order $[k,\ldots,d,1,\ldots,k-1]$, and then performing matricization along the first l modes.

Theorem 3 [66], [67]: Suppose that $\mathcal{A}\in\mathbb{R}^{n_1\times\cdots\times n_d}$ is an order-d tensor with TR rank $[r_1,\ldots,r_d]$, then for each TR unfolding matrix $\mathcal{A}^{\langle k,l\rangle}$, we have

$$\mathrm{rank}(\mathcal{A}^{\langle k,l\rangle})\le r_k r_{k+l}$$

with $r_i=r_{i+d}$, $i\in\{1,2,\ldots,d\}$, where $\mathrm{rank}(\cdot)$ denotes the rank of a matrix.

Definition 4 (TRNN [67]): Assume that the tensor $\mathcal{A}\in\mathbb{R}^{n_1\times\cdots\times n_d}$ with TR-form, then its TR nuclear norm (TRNN) is defined by

$$\|\mathcal{A}\|_{\mathrm{trnn}}:=\sum_{k=1}^{d}w_k\cdot\left\|\mathcal{A}^{\langle k,l\rangle}\right\|_{\circledast}=\sum_{k=1}^{d}w_k\cdot\sum_{i}\sigma_i\left(\mathcal{A}^{\langle k,l\rangle}\right)$$

where $\{w_k\}_{k=1}^{d}$ are the non-negative weights with $\sum_{k=1}^{d}w_k=1$, and $\sigma_i(\cdot)$ is denoted as the ith singular value of a matrix. Note that the literature [66], [68] states that $\lceil d/2\rceil$ unfoldings are sufficient for low-rank TR modeling.

III. Generalized Nonconvex Approach for HAD

This section introduces a generalized nonconvex framework for the HAD task, organized into four sections for detailed elaboration. Section III-A devises two novel regularizers to respectively characterize the HSI's background and anomaly components. On this basis, an effective generalized nonconvex HAD model is proposed in Section III-B. Section III-C provides solution strategies for pivotal subproblems, based on which an efficient ADMM algorithm is developed to solve the formulated model in Section III-D.

A. Generalized Nonconvex Regularizers

Let HSI data be presented by a third-order tensor $\mathcal{M}\in\mathbb{R}^{n_1\times n_2\times n_3}$, where n_3 indicates the amount of spectral bands, and n_1 and n_2 are the height and width of each band, respectively. We decompose raw HSI $\mathcal{M}$ into a background tensor $\mathcal{L}$ and an anomaly tensor $\mathcal{E}$. Then, we can solve the HAD problem through the following tensor RPCA model:

$$\min_{\mathcal{L},\mathcal{E}}\ \Psi(\mathcal{L})+\lambda\cdot\Upsilon(\mathcal{E}),\quad \text{s.t. } \mathcal{L}+\mathcal{E}=\mathcal{M}\tag{1}$$

where $\Psi(\mathcal{L})$ represents the regularizer measuring the prior structures of HSI's background component, $\Upsilon(\mathcal{E})$ denotes the

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anomaly-sparsity regularization, and $\lambda > 0$ is a trade-off parameter that balances these two terms.

Next, it is important to design the above two regularizers to effectively encode the prior structures of HSI data.

1) Novel Regularizer Encoding Prior Structures of Background Tensor: Leveraging gradient map modeling [50], [59], [60], [61], [62], circular TR unfolding [66], [67], [68], [69], [70] and nonconvex regularization techniques [71], [72], [73], [74], [75], [76], we put forward a generalized and efficient regularizer to encode the $\mathbf{L} + \mathbf{S}$ priors of the background tensor. The motivation for designing this regularizer comes from the following two aspects. On the one hand, motivated by the fact that $\mathbf{L}$ and $\mathbf{S}$ priors always coexist with each other in HSI data [59], [60], [61], we consider proposing a novel regularization penalty which can simultaneously characterize these two types of priors with a unique term under the TR framework. This single regularizer can promote the two priors concurrently, and overcome the difficulty of tuning the trade-off parameter imposed between these two regularizers. On the other hand, the low-rank regularizer modeled by the TRNN [66], [67] equally treats all singular components of each TR unfolding matrix, thus potentially causing some unavoidable biases in practical applications. To alleviate this weakness, although Ding et al. [69] put forward an enhanced low-rank measure scheme, it still lacks a unified nonconvex form. Besides, these studies under the TR framework do not consider other critical prior structures underlying tensor data, e.g., $\mathbf{S}$ prior. Therefore, to better encode the $\mathbf{L}$ and $\mathbf{S}$ priors of the background of HSI data simultaneously, we propose a unified nonconvex regularizer involving a family of effective penalty functions within the TR framework.

Before formally introducing the newly devised regularizer, we first give the definition of gradient tensor (also called gradient map).

Definition 5 (Gradient Tensor [61]): For $\mathcal{A} \in \mathbb{R}^{n_1 \times \cdots \times n_d}$, its gradient tensor along the tth mode is defined as follows:

$$\mathcal{G}_t := \nabla_t(\mathcal{A}) = \mathcal{A} \times_t \boldsymbol{D}_{n_t}, \quad t = 1, 2, \ldots, d \tag{2}$$

where $\boldsymbol{D}_{n_t}$ is a row circulant matrix of $(-1, 1, 0, \ldots, 0)$, ∇_t is defined as the corresponding difference operator along the tth mode of tensor $\mathcal{A}$, and $\mathcal{A} \times_t \boldsymbol{D}_{n_t}$ denotes the mode-t product of tensor $\mathcal{A}$ with matrix $\boldsymbol{D}_{n_t}$.

Definition 6 (GNTCTV): For a background tensor $\mathcal{L} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, let $\mathcal{G}_t$ denote the gradient tensor along the tth mode of $\mathcal{L}$. Then, the generalized nonconvex TCTV (GNTCTV) norm of $\mathcal{L}$ is defined as follows:

$$\begin{aligned}\Psi(\mathcal{L}) &:= \frac{1}{3}\sum_{t=1}^{3}\left\{\sum_{k=1}^{2} w_k \cdot \Phi\left(\mathcal{G}_t^{\langle k,l\rangle}\right)\right\} \\ &= \frac{1}{3}\sum_{t=1}^{3}\left\{\sum_{k=1}^{2} w_k \cdot \sum_{i} \Phi\left(\sigma_i\left(\mathcal{G}_t{}^{\langle k,l\rangle}\right)\right)\right\}\end{aligned} \tag{3}$$

where $\{w_k\}_{k=1}^2$ are the non-negative weights with $\sum_k w_k = 1$, and $\Phi(\cdot) : \mathbb{R}^+ \to \mathbb{R}^+$ is a generalized nonconvex function, which satisfies the following assumptions.

1) $\Phi(\cdot)$: $\mathbb{R}^+ \to \mathbb{R}^+$ is proper, lower semicontinuous, and symmetric with respect to the y-axis.
2) $\Phi(\cdot)$ is concave and monotonically increasing on $[0, \infty)$ with $\Phi(0) = 0$.

Many popular nonconvex penalty functions satisfy the above assumptions, such as firm function [77], logarithmic (Log) function [78], ℓ_q function [79], smoothly clipped absolute deviation (SCAD) function [80], and minimax concave penalty (MCP) function [81]. Thus, in the above Definition 6, we effectively employ a family of nonconvex functions on the singular values of all circular TR unfolding matrices derived from each gradient tensor.

2) Novel Anomaly-Sparsity Regularizer: In order to detect the sparse anomaly component existing in HSI data more robustly, we define the following generalized nonconvex regularization penalty, i.e.,

$$\Upsilon(\mathcal{E}) := \psi\left(h(\mathcal{E})\right) \tag{4}$$

where $\psi(\cdot)$ has the same properties as $\Phi(\cdot)$. Here, two types of sparse anomalies are taken into account. When the anomaly tensor $\mathcal{E}$ has structured sparsity on the tubes, $h(\cdot) = \|\cdot\|_{F,1}$ is defined as an $\ell_{F,1}$-norm. Then, we have

$$\Upsilon(\mathcal{E}) := \psi(\|\mathcal{E}\|_{F,1}) = \sum_{i_1=1}^{n_1}\sum_{i_2=1}^{n_2}\psi\left(\|\mathcal{E}_{i_1 i_2 :}\|_F\right). \tag{5}$$

When the tensor $\mathcal{E}$ is an entrywise anomaly tensor, $h(\cdot) = \|\cdot\|_1$ is defined as an ℓ_1-norm. Then, we have

$$\Upsilon(\mathcal{E}) := \psi(\|\mathcal{E}\|_1) = \sum_{i_1=1}^{n_1}\sum_{i_2=1}^{n_2}\sum_{i_3=1}^{n_3}\psi\left(|\mathcal{E}_{i_1 i_2 i_3}|\right). \tag{6}$$

B. Generalized Nonconvex Model

On the basis of Section III-A, this section proposes a TR framework-based generalized nonconvex HAD model (abbreviated as TR-GNHAD), i.e.,

$$\begin{aligned}&\min_{\mathcal{L},\mathcal{E}} \ \frac{1}{3}\sum_{t=1}^{3}\left\{\sum_{k=1}^{2} w_k \cdot \Phi\left(\mathcal{G}_t^{\langle k,l\rangle}\right)\right\} + \lambda \cdot \psi\left(h(\mathcal{E})\right) \\ &\text{s.t. } \mathcal{G}_t = \nabla_t(\mathcal{L}), \quad \mathcal{L} + \mathcal{E} = \mathcal{M}.\end{aligned} \tag{7}$$

The advantages of our model (7) are summarized as follows.

1) Compared with the existing tensor-based HAD models (e.g., [37], [38], [39], [40], [41], [42], [43], [48], [49], [50], [51], [52]), the formulated one possesses a more generalized and effective form. Specifically, two novel generalized regularizers $\Psi(\cdot)$ and $\Upsilon(\cdot)$ involving a family of nonconvex functions are integrated into the proposed HAD model. The former can synchronously encode both $\mathbf{L}$ and $\mathbf{S}$ prior structures of the background of HSI data. This strategy eliminates the hassle of tuning the trade-off parameter imposed between these two regularizers. The latter has a superior potential to be sparsity-promoting in comparison with the commonly used ℓ_1-norm and $\ell_{F,1}$-norm, and thus it can well detect and locate the abnormal component that existed in the HSI data.
2) The existing matrix-based HAD methods (e.g., [34], [35], [36], [45], [46], [47]) are limited to the case of matrix data. These approaches require directly reshaping

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the raw HSI data into the matrix form when we perform the HAD tasks. However, this process destroys the intrinsic structure of the original HSI data, thus resulting in suboptimal detection performance. On the contrary, our method takes into account the circular TR unfolding strategy, which can well retain the internal structural information of HSI data, and fully exploit its spatial and spectral correlations.

C. Generalized Solver for Key Subproblems

This section mainly presents the solution methods of two kinds of key subproblems involved in our model (7), i.e.,

$$\arg\min_{\mathcal{E}} \ \lambda \cdot \psi\left(h(\mathcal{E})\right) + \frac{1}{2}\|\mathcal{E}-\mathcal{A}\|_F^2 \tag{8}$$

$$\arg\min_{\mathcal{L}} \ \tau \cdot \Phi(\mathcal{L}^{\langle k,l\rangle}) + \frac{1}{2}\|\mathcal{L}-\mathcal{A}\|_F^2. \tag{9}$$

Before providing the solutions to the problems (8) and (9), we first introduce the proximal mappings of the generalized nonconvex functions, which play a central role in developing highly efficient first-order algorithms. Specifically, for a nonconvex penalty function $\psi(\cdot)$ satisfying the assumptions listed in Definition 6, its proximity operator is defined as follows:

$$\text{Prox}_{\psi,\mu}(v) = \arg\min_{x} \left\{\mu \cdot \psi(x) + \frac{1}{2}(x-v)^2\right\} \tag{10}$$

where $\mu > 0$ is a penalty parameter. Below, we summarize the proximity operators for several popular nonconvex regularization penalties, including firm thresholding, ℓ_q thresholding, MCP, Log, and SCAD penalties.

1) *Firm Thresholding [77]:* The firm penalty is given by

$$\psi(x) = \begin{cases} a\left(|x| - \dfrac{x^2}{2\gamma}\right), & |x| < \gamma \\ \dfrac{a\gamma}{2}, & |x| \geq \gamma \end{cases}$$

where $a, \gamma > 0$, and

$$\text{Prox}_{\psi,\mu}(v)\begin{cases} 0, & |v| \leq a\mu \\ \dfrac{\text{sign}(v)(|v|-a\mu)\gamma}{\gamma - a\mu}, & a\mu < |v| \leq \gamma \\ v, & |v| > \gamma \end{cases}$$

where $\gamma > a\mu$ and $\text{sign}(\cdot)$ denotes the signum function.

2) *SCAD [80]:* The SCAD penalty is given by

$$\psi(x) = \begin{cases} \xi|x|, & |x| < \xi \\ \dfrac{2b\xi|x| - x^2 - \xi^2}{2(b-1)}, & \xi \leq |x| < b\xi \\ \dfrac{(b+1)\xi^2}{2}, & |x| \geq b\xi \end{cases}$$

where $b \geq 1, \xi > 0$, and for any $0 < \mu < b-1$

$$\text{Prox}_{\psi,\mu}(v)\begin{cases} \text{sign}(v)\max(|v| - \xi\mu, 0) \\ \quad |v| \leq (1+\mu)\xi \\ \dfrac{(b-1)v - \text{sign}(v)b\mu\xi}{b-1-\mu} \\ \quad (1+\mu)\xi \leq |v| < b\xi \\ v, \quad |v| \geq b\xi. \end{cases}$$

3) *MCP [81]:* The MCP function is defined by

$$\psi(x) = \begin{cases} c|x| - \dfrac{x^2}{2\eta}, & |x| \leq c\eta \\ \dfrac{c^2\eta}{2}, & |x| > c\eta \end{cases}$$

where $c, \eta > 0$, and for any $0 < \mu < \eta$

$$\text{Prox}_{\psi,\mu}(v)\begin{cases} 0, & |v| \leq c\mu \\ \dfrac{\text{sign}(v)(|v| - c\mu)}{1-\mu/\eta}, & c\mu < |v| \leq c\eta \\ v, & |v| > c\eta. \end{cases}$$

4) *ℓ_q Thresholding [79]:* The penalty function is defined by $\psi(x) = |x|^q$, where $0 < q < 1$. Except for two known cases of $q = 1/2$ and $q = 2/3$, ℓ_q thresholding does not have a closed-form expression, but it can be solved by

$$\text{Prox}_{\psi,\mu}(v) = \begin{cases} 0, & |v| < \delta \\ \{0, \text{sign}(v)\hat{\alpha}\}, & |v| = \delta \\ \text{sign}(v)\hat{\alpha}^*, & |v| > \delta \end{cases}$$

where $\hat{\alpha} = [2\mu(1-q)]^{(1/(2-q))}$, $\delta = \hat{\alpha} + \mu q\hat{\alpha}^{q-1}$, and $\hat{\alpha}^*$ is the largest solution of $\alpha + \mu q\alpha^{q-1} - |v| = 0$, where $\alpha > 0$. With the initial value $\alpha_{(0)} \in (\hat{\alpha}, |v|)$, $\hat{\alpha}^*$ can be obtained from the iteration

$$\alpha_{(j+1)} = |v| - \mu q(\alpha_{(j)})^{q-1}, \quad j = 0, 1, 2, \ldots$$

5) *Log [78]:* The logarithmic function is defined by $\psi(x) = \log(1 + (|x|/\theta))$, where $\theta > 0$. Then,

$$\text{Prox}_{\psi,\mu}(v) = \text{sign}(v)z$$

where z is an optimal solution of the problem $z = \arg\min_{x\in\Theta}\{\mu\psi(x) + (1/2)(x-v)^2\}$, and Θ is a set composed of 3 elements or 1 element. If $a := (|v|-\theta)^2 - 4(\mu - \theta|v|) \geq 0$

$$\Theta = \left\{0, \max\left\{\frac{|v| - \theta + \sqrt{a}}{2}, 0\right\}, \max\left\{\frac{|v| - \theta - \sqrt{a}}{2}, 0\right\}\right\}$$

otherwise, $\Theta = \{0\}$.

1) Solve the Problem (8) via Generalized Nonconvex Thresholding/Shrinkage Operator: Based on the previous analysis, the optimal solution to the subproblem (8) can be computed by generalized nonconvex thresholding/shrinkage operator in an elementwise or tubewise manner. Please see Algorithm 1 for details.

2) Solve the Problem (9) via GNSVT Operator: The closed-form solution to the problem (9) can be obtained via the following key Theorem 7, i.e.,

$$\mathcal{L} = \text{fold}_{(k,l)}\left(\mathfrak{D}_{\Phi,\tau}(\mathcal{A}^{\langle k,l\rangle})\right)$$

where $\text{fold}_{(k,l)}(\cdot)$ denotes the inverse operator of k-shifting l-matricization which satisfies $\text{fold}_{(k,l)}(\mathcal{A}^{\langle k,l\rangle}) = \mathcal{A}$.

Theorem 7 (GNSVT [82]): Let the singular value decomposition (SVD) of $\boldsymbol{Y} \in \mathbb{R}^{n_1\times n_2}$ be $\boldsymbol{Y} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T$, $m =$

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Algorithm 1 Generalized Thresholding Operator

Input: $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, $\lambda > 0$, $\psi(\cdot)$.

```
if h(·) = ‖·‖_1 then
    Ê_{i1i2i3} = Prox_{ψ,λ}(A_{i1i2i3});
end
else if h(·) = ‖·‖_{F,1} then
    for i = 1, 2, ..., n1 do
        for j = 1, 2, ..., n2 do
            Ê_{ij:} = A_{ij:}/‖A_{ij:}‖_F · Prox_{ψ,λ}(‖A_{ij:}‖_F);
        end
    end
end
```

Output: $\hat{\mathcal{E}} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$.

$\min(n_1, n_2)$. For any $\tau \geq 0$, then a global optimal solution to the following optimization problem

$$\min_{X} \ \tau \cdot \Phi(X) + \frac{1}{2}\|X - Y\|_F^2$$

is given by

$$\mathcal{D}_{\Phi,\tau}(Y) = U \cdot \tilde{Y} \cdot V^T$$

where $\tilde{Y} = \text{diag}(\phi_1, \phi_2, \ldots, \phi_m)$, $\phi_i = \text{Prox}_{\Phi,\tau}(\Sigma_{i,i})$, $i \in \{1, 2, \ldots, m\}$, and $\text{Prox}_{\Phi,\tau}(\cdot)$ denotes the proximity operator of nonconvex penalty function $\Phi(\cdot)$, which has the same properties as $\psi(\cdot)$.

D. Optimization Algorithm

In this section, the ADMM framework [64] is adopted to solve the proposed model (7). The nonconvex model (7) can be equivalently reformulated as follows:

$$\begin{aligned} \min_{\mathcal{L},\mathcal{E},\mathcal{G}_t^{(k)}} \ & \frac{1}{3}\sum_{t=1}^{3}\left\{\sum_{k=1}^{2} w_k \cdot \Phi\left(\left(\mathcal{G}_t^{(k)}\right)^{\langle k,l\rangle}\right)\right\} + \lambda \cdot \psi\left(h(\mathcal{E})\right) \\ \text{s.t.} \ & \mathcal{G}_t^{(k)} = \nabla_t(\mathcal{L}), \quad \mathcal{L} + \mathcal{E} = \mathcal{M}. \end{aligned} \tag{11}$$

The augmented Lagrangian function of (11) is

$$\begin{aligned} &\mathcal{F}\left(\mathcal{L}, \mathcal{G}_t^{(k)}, \Lambda_t^{(k)}, \mathcal{E}, \mathcal{Y}\right) \\ &= \frac{1}{3}\sum_{t=1}^{3}\sum_{k=1}^{2}\Big\{w_k \cdot \Phi\left(\left(\mathcal{G}_t^{(k)}\right)^{\langle k,l\rangle}\right) + \left\langle \Lambda_t^{(k)}, \nabla_t(\mathcal{L}) - \mathcal{G}_t^{(k)}\right\rangle \\ &\qquad + \frac{\mu}{2}\left\|\nabla_t(\mathcal{L}) - \mathcal{G}_t^{(k)}\right\|_F^2\Big\} \\ &\quad + \lambda \cdot \psi\left(h(\mathcal{E})\right) + \langle \mathcal{Y}, \mathcal{M} - \mathcal{L} - \mathcal{E}\rangle \\ &\quad + \frac{\mu}{2}\|\mathcal{M} - \mathcal{L} - \mathcal{E}\|_F^2 \end{aligned} \tag{12}$$

where μ is the regularization parameter, and $\{\Lambda_t^{(k)}\}$ and $\mathcal{Y}$ are Lagrange multipliers. It can be further expressed as follows:

$$\begin{aligned} &\mathcal{F}\left(\mathcal{L}, \mathcal{G}_t^{(k)}, \Lambda_t^{(k)}, \mathcal{E}, \mathcal{Y}\right) \\ &= \frac{1}{3}\sum_{t=1}^{3}\sum_{k=1}^{2}\Big\{w_k \cdot \Phi\left(\left(\mathcal{G}_t^{(k)}\right)^{\langle k,l\rangle}\right) \\ &\qquad + \frac{\mu}{2}\left\|\nabla_t(\mathcal{L}) - \mathcal{G}_t^{(k)} + \Lambda_t^{(k)}/\mu\right\|_F^2\Big\} \\ &\quad + \lambda \cdot \psi\left(h(\mathcal{E})\right) + \frac{\mu}{2}\|\mathcal{M} - \mathcal{L} - \mathcal{E} + \mathcal{Y}/\mu\|_F^2 + C \end{aligned} \tag{13}$$

where C is only the multiplier-dependent squared item. Below, we show how to solve the subproblems for each involved variable.

1) Update $\mathcal{L}^{\{v+1\}}$: Taking the derivative in (13) with respect to $\mathcal{L}$, it gets the following linear system:

$$\begin{aligned} &\left(\mathcal{I} + \sum_{t=1}^{3}\sum_{k=1}^{2}\nabla_t^T\nabla_t\right)(\mathcal{L}) \\ &\quad = \left(\mathcal{M} - \mathcal{E}^{\{v\}} + \frac{\mathcal{Y}^{\{v\}}}{\mu^{\{v\}}}\right) \\ &\qquad + \sum_{t=1}^{3}\sum_{k=1}^{2}\nabla_t^T\left(\left(\mathcal{G}_t^{(k)}\right)^{\{v\}} - \frac{\left(\Lambda_t^{(k)}\right)^{\{v\}}}{\mu^{\{v\}}}\right) \end{aligned} \tag{14}$$

where $\nabla_t^T(\cdot)$ denotes the transpose operator of $\nabla_t(\cdot)$. Referring to the literature [83], we can apply multidimensional FFT, which diagonalizes $\nabla_t(\cdot)$'s corresponding difference tensors $\mathcal{D}_t$, enabling to efficiently obtain the optimal solution of (14) via the convolution theorem of Fourier transforms, i.e.,

$$\mathcal{L}^{\{v+1\}} = \mathscr{F}^{-1}\left(\frac{\mathscr{F}\left(\mathcal{M} - \mathcal{E}^{\{v\}} + \frac{\mathcal{Y}^{\{v\}}}{\mu^{\{v\}}}\right) + \mathcal{T}}{\mathbf{1} + \sum_{t=1}^{3}\sum_{k=1}^{2}\mathscr{F}(\mathcal{D}_t)^* \odot \mathscr{F}(\mathcal{D}_t)}\right)$$

where $\mathcal{T} = \sum_{t=1}^{3}\sum_{k=1}^{2}\mathscr{F}(\mathcal{D}_t)^* \odot \mathscr{F}((\mathcal{G}_t^{(k)})^{\{v\}} - (((\Lambda_t^{(k)})^{\{v\}})/\mu^{\{v\}}))$, $\mathbf{1}$ is a tensor with all elements as 1, $\odot$ is the componentwise multiplication, and $\mathscr{F}(\cdot)$ and $\mathscr{F}^{-1}(\cdot)$ are the FFT and its inverse transform, respectively.

2) Update $(\mathcal{G}_t^{(k)})^{\{v+1\}}$: For each $t = 1, 2, 3$, $k = 1, 2$, extracting all items containing $\mathcal{G}_t^{(k)}$ from (12), we have

$$\arg\min_{\mathcal{G}_t^{(k)}} \ w_k \cdot \Phi\left(\left(\mathcal{G}_t^{(k)}\right)^{\langle k,l\rangle}\right) + \frac{\mu^{\{v\}}}{2}\left\|\mathcal{K} - \mathcal{G}_t^{(k)}\right\|_F^2. \tag{15}$$

where $\mathcal{K} = \nabla_t(\mathcal{L}^{\{v+1\}}) + (((\Lambda_t^{(k)})^{\{v\}})/\mu^{\{v\}})$. This key subproblem is analogous to the form of the minimization problem (9), and its closed-form solution can be obtained by utilizing the generalized nonconvex singular value thresholding (GNSVT) operation (see III-C for details), i.e.,

$$\left(\mathcal{G}_t^{(k)}\right)^{\{v+1\}} = \text{fold}_{\langle k,l\rangle}\left(\mathcal{D}_{\Phi,\frac{w_k}{\mu^{\{v\}}}}(\mathcal{K}^{\langle k,l\rangle})\right).$$

Note that the adjustment of the weight w_k is similar to the previous work [68], and l is set to be 2 by default.

3) Update $\mathcal{E}^{\{v+1\}}$: The optimization subproblem with respect to $\mathcal{E}$ can be written as follows:

$$\mathcal{E}^{\{v+1\}} = \arg\min_{\mathcal{E}} \ \lambda \cdot \psi\left(h(\mathcal{E})\right) + \frac{\mu^{\{v\}}}{2}\left\|\mathcal{E} - \mathcal{H}\right\|_F^2 \tag{16}$$

where $\mathcal{H} = \mathcal{M} - \mathcal{L}^{\{v+1\}} + (\mathcal{Y}^{\{v\}}/\mu^{\{v\}})$. This subproblem is equivalent to the form of the minimization problem (8), and its closed-form solution can be obtained by utilizing the generalized nonconvex shrinkage operator. Please see Section III-C for details.

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Algorithm 2 Solve the Proposed TR-GNHAD Model (7) via ADMM Framework

Input: HSI data: $\mathcal{M}$, $\Phi(\cdot)$, $\psi(\cdot)$, $h(\cdot)$, λ.
Initialize: $(\mathcal{G}_t^{(k)})^{\{0\}} = \mathcal{E}^{\{0\}} = (\Lambda_t^{(k)})^{\{0\}} = \mathcal{Y}^{\{0\}} = \mathbf{0}$, ϑ, $\mu^{\{0\}}$, $\mu^{\max}$, ϖ, $\nu = 0$;
while *not converged* **do**
  Update $\mathcal{L}^{\{\nu+1\}}$ by computing (14) ;
  Update $(\mathcal{G}_t^{(k)})^{\{\nu+1\}}$ by computing (15);
  Update $\mathcal{E}^{\{\nu+1\}}$ by computing (16);
  Update $(\Lambda_t^{(k)})^{\{\nu+1\}}$, $\mathcal{Y}^{\{\nu+1\}}$, $\mu^{\{\nu+1\}}$ by (17)-(19);
  Update the detection map $\boldsymbol{T}^{\{\nu+1\}}$ by computing (20);
  Check the convergence conditions by computing (21);
end
Output: background $\mathcal{L}$, anomalies $\mathcal{E}$, detection map $\boldsymbol{T}$.

4) Update $(\Lambda_t^{(k)})^{\{\nu+1\}}$, $\mathcal{Y}^{\{\nu+1\}}$, and $\mu^{\{\nu+1\}}$ (Lagrange Multipliers and Penalty Parameter): Based on the rule of ADMM framework, the Lagrange multipliers are updated by the following equations:

$$\mathcal{Y}^{\{\nu+1\}} = \mathcal{Y}^{\{\nu\}} + \mu^{\{\nu\}}(\mathcal{M} - \mathcal{L}^{\{\nu+1\}} - \mathcal{E}^{\{\nu+1\}}) \tag{17}$$

$$\left(\Lambda_t^{(k)}\right)^{\{\nu+1\}} = \left(\Lambda_t^{(k)}\right)^{\{\nu\}} + \mu^{\{\nu\}} \cdot \mathcal{N} \tag{18}$$

$$\mu^{\{\nu+1\}} = \min\left(\mu^{\max}, \vartheta\mu^{\{\nu\}}\right) \tag{19}$$

where $\mathcal{N} = \nabla_t(\mathcal{L}^{\{\nu+1\}}) - (\mathcal{G}_t^{(k)})^{\{\nu+1\}}$, and ϑ stands for the growth rate.

5) Update the Detection Map $\boldsymbol{T}^{\{\nu+1\}}$: Finally, the anomaly detection map can be obtained by

$$\boldsymbol{T}^{\{\nu+1\}}(i_1, i_2) = \sqrt{\sum_{i_3=1}^{n_3} |\mathcal{E}^{\{\nu+1\}}(i_1, i_2, i_3)|^2}. \tag{20}$$

6) Check the Convergence Condition: The convergence condition is defined as follows:

$$\max\left\{\begin{array}{l} \|\mathcal{L}^{\{\nu+1\}} - \mathcal{L}^{\{\nu\}}\|_\infty, \|\mathcal{M} - \mathcal{L}^{\{\nu+1\}} - \mathcal{E}^{\{\nu+1\}}\|_\infty \\ \|\mathcal{E}^{\{\nu+1\}} - \mathcal{E}^{\{\nu\}}\|_\infty, \left\|\left(\mathcal{G}_t^{(k)}\right)^{\{\nu+1\}} - \left(\mathcal{G}_t^{(k)}\right)^{\{\nu\}}\right\|_\infty \\ \left\|\nabla_t(\mathcal{L}^{\{\nu+1\}}) - \left(\mathcal{G}_t^{(k)}\right)^{\{\nu+1\}}\right\|_\infty \end{array}\right\} \leq \varpi. \tag{21}$$

E. Complexity Analysis

Given an input HSI data with the size of $n_1 \times n_2 \times n_3$, we analyze the per-iteration complexity of Algorithm 2, which needs to update $\mathcal{L}$, $\mathcal{G}_t^{(k)}$, $\mathcal{E}$, $\Lambda_t^{(k)}$, and $\mathcal{Y}$, respectively. For simplicity, we set $n_1 = n_2 = n_3 = N$. Specifically, we observe the following: 1) the time complexity of calculating $\mathcal{L}$ that mainly uses FFT is $\mathcal{O}(N^3 \log N^3)$; 2) The update of $\mathcal{G}_t^{(k)}$, which mainly involves the GNSVT operation of circular TR unfolding matrices of all gradient tensors, equals to $\mathcal{O}(N^4)$; 3) Upadating $\mathcal{E}$ requires to perform generalized thresholding operation with a complexity of $\mathcal{O}(N^3)$; and 4) $\mathcal{Y}$ and $\Lambda_t^{(k)}$ can be updated by a low consumed algebraic computation.

IV. Experimental Results

In this section, we perform extensive experiments on several HSI datasets to substantiate the practicability, effectiveness, and superiority of the proposed TR-GNHAD approach. All the experiments are run on the following platforms: 1) Windows 10 and MATLAB (R2016a) with an Intel[1] Xeon[1] Gold-5122 3.60 GHz CPU and 192 GB memory and 2) Windows 10 and MATLAB (R2022b) with an Intel[1] Xeon[1] Gold-6230 2.10 GHz CPU and 128 GB memory.

A. Experimental Datasets

1) Salinas Dataset [31]: This simulated dataset with the size of 120 × 120 × 204 is provided by Tan et al., which is crafted based on the Salinas scene. This scene was captured by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor from Salinas Valley, CA, USA.

2) Pavia-Center Dataset [33]: The dataset with the size of 105 × 120 × 102 was expertly gathered by the reflective optics system imaging spectrometer (ROSIS) at the Pavia Center, Italy. The background comprises bridge and river, while the vehicles on the bridge and the exposed soil adjacent to the piers are classified as anomalous targets.

3) Hyperion Dataset [84], [85]: This dataset with the size of 150 × 150 × 155 was acquired by the Hyperion imaging sensor, encompassing an agricultural region of the State of Indiana, USA, in 2008. In this dataset, the anomalies originate from the storage silos and the roofs, which are visually indistinguishable from the background.

4) HYDICE Dataset [23]: The dataset with the size of 100 × 80 × 162 was shot by the hyperspectral digital imagery collection experiment (HYDICE) airborne sensor over an urban area in CA, USA. The background of the image primarily comprises a parking lot, water, land, and two roads. The vehicles and roofs are regarded as anomalies and their pixels constitute approximately 0.24% of the entire scene.

5) Segundo Dataset [15]: This dataset with the size of 250 × 300 × 224 was captured by the AVIRIS sensor, which covers an area of El Segundo, CA, USA. The size of each spectral bands is resized to 100 × 120 in our experiment. The HSI dataset predominantly encompasses an area of oil refinery, several residential areas, parks, and one school zone. The constructions of oil refineries such as towers and storage tanks are identified as anomalies.

6) Airport Datasets[2] [86]: These datasets were manually obtained by the AVIRIS over the regions situated in Los Angeles and Gulfport, USA. Objects located on the Earth's surface, including roads and buildings, are viewed as the background. Conversely, anomalies are represented by flights of different heights. We mainly select Airport-1 (100 × 100 × 205) and Airport-4 (100 × 100 × 191) as the tested HSI data.

7) Urban Datasets [86]: The Urban datasets were collected with AVIRIS sensors and contained five HSI data located in different scenes. Roofs and other objects that are notably distinct from their surrounding environments will be perceived

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[2]https://ehu.eus/ccwintco/index.php?title = Hyperspectral_Remote_Sensing_Scenes

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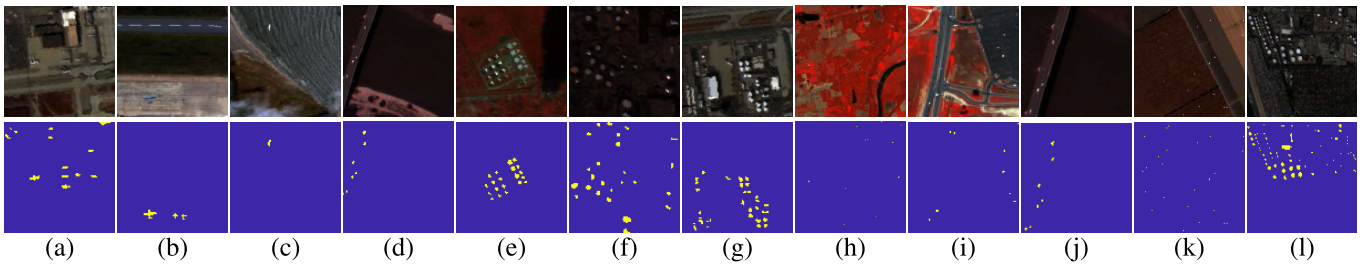


Fig. 3. Pseudocolor images and ground-truth maps of 12 HSI datasets. (a) Airport-1. (b) Airport-4. (c) Beach-3. (d) Beach-4. (e) Urban-2. (f) Urban-4. (g) Urban-5. (h) Hyperion. (i) HYDICE. (j) Pavia. (k) Salinas. (l) Segundo.

as anomalies. We select Urban-4, Urban-5 ($100 \times 100 \times 205$), and Urban-2 ($100 \times 100 \times 207$) as the tested HSI data.

8) Beach Datasets [86]: In these datasets, ships, vehicles, flights, and man-made objects are labeled as anomalies, whereas beaches and water are designated as the backgrounds. We mainly select Beach-3 ($100 \times 100 \times 188$) and Beach-4 ($150 \times 150 \times 102$) as the tested HSI data.

The pseudocolor map and the corresponding ground-truth map of the aforementioned HSI datasets are shown in Fig. 3. In our experiments, each raw HSI data is conducted with band-by-band normalization operation.

B. Comparative Experiments

1) Comparison Methods: To showcase the effectiveness of the proposed HAD algorithm, some baselines are introduced as compared detectors in our experiments, which chiefly contain one classical static-based method: RX [17], one collaborative representation-based method: CRD [30], one deep learning-based method: DeCNN-AD [24], three matrix-based methods: LSMAD [46], LRASR [34], and GTVLRR [35], and five tensor-based methods: PTA [48], TPCA [49], PCA-TLRSR [37], GNLTR [39], and TRDFTVAD [51].

2) Parameters Setting: Related parameters for all competing algorithms are selected from a grid search, so that each anomaly detection method is optimized for a fair comparison. Specifically, **RX** and **TPCA** are parameter-free. The parameters of window size $(w_{\text{in}}, w_{\text{out}})$ in **CRD** algorithm are selected from $\{(3,5), (5,7), (7,9), (11,13), (15,17), (17,19), (19,21)\}$. Besides, the regularization parameter λ is chosen from $\{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$. With regard to **DeCNN-AD** algorithm, K varies from 1 to 20, and the values of β and λ change from 10^{-4} to 1. For **LSMAD** algorithm, the empirical rank value is taken as 5∼20. Two parameters β and λ in **LRASR** algorithm are chosen from $[0.001, 0.01, 0.1, 0.5, 1, 2, 3]$ and $[0.001, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5]$, respectively. There are three regularization parameters in **GTVLRR** method, that is, λ, β, and γ. We determine the value of λ and β from $\{0.005, 0.05, 0.1, 0.3, 0.5, 0.7, 1\}$ and γ from $\{0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.5\}$. For the background dictionary in **LRASR** and **GTVLRR** algorithms, the number of clusters is set to 15, and 20 pixels are selected in each cluster as atoms. In the **PTA** method, the truncated rank value r is selected from 0∼20 and the regularization parameter μ is selected from $\{10^{-4}, 10^{-3}, 10^{-2}, 0.1, 1, 10, 10^2, 10^3, 10^4\}$. Other parameters of the **PTA** algorithm remain unchanged as in [48]. For the **PCA-TLRSR**, the parameters λ and λ' are chosen from $\{0.001, 0.005, 0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5\}$. The optimal combinations of two parameters λ and β in **TRDFTVAD**, are both searched from $\{0.01, 0.05, 0.1, 0.5, 1, 5, 10\}$, where TR rank is fixed as $(r_1, r_2, r_3) = (5, 5, 5)$. The hyperparameters θ and α involved in **GNLTR** method are selected from $\{0.0005, 0.001, 0.01, 0.03, 0.05, 0.07, 0.1\}$, and the nonconvex surrogate is chosen as ℓ_q-norm. Note that PCA technique is adopted to preprocess the original HSI data in **GNLTR** and **PCA-TLRSR** methods.

3) Evaluation Metrics: To effectively evaluate the performance of all HAD detectors, we introduce the 3-D receiver operating characteristic (ROC) curve $(\mathrm{P}_D, \mathrm{P}_F, \tau)$ [87], and its induced three 2-D ROC curves of $(\mathrm{P}_D, \mathrm{P}_F)$, (P_D, τ), and (P_F, τ) with their corresponding area under curve (AUC) values that are denoted as $\mathrm{AUC}_{(\mathrm{P}_D,\mathrm{P}_F)}$, $\mathrm{AUC}_{(\mathrm{P}_D,\tau)}$, and $\mathrm{AUC}_{(\mathrm{P}_F,\tau)}$, respectively. In addition, we also employ other AUC evaluation metrics [87] based on the 3-D ROC curve to comprehensively assess the detection performance of the HAD detector. These AUC evaluation metrics mainly include $\mathrm{AUC}_{(\mathrm{ODP})}$, $\mathrm{AUC}_{(\mathrm{SNPR})}$, and $\mathrm{AUC}_{(\mathrm{TDBS})}$, which are defined as follows:

$$\begin{cases} \mathrm{AUC}_{(\mathrm{ODP})} = \mathrm{AUC}_{(\mathrm{P}_D,\mathrm{P}_F)} + \mathrm{AUC}_{(\mathrm{P}_D,\tau)} - \mathrm{AUC}_{(\mathrm{P}_F,\tau)} \\ \mathrm{AUC}_{(\mathrm{SNPR})} = \mathrm{AUC}_{(\mathrm{P}_D,\tau)} / \mathrm{AUC}_{(\mathrm{P}_F,\tau)} \\ \mathrm{AUC}_{(\mathrm{TDBS})} = \mathrm{AUC}_{(\mathrm{P}_D,\tau)} - \mathrm{AUC}_{(\mathrm{P}_F,\tau)}. \end{cases}$$

The evaluation indicators $\mathrm{AUC}_{(\mathrm{P}_D,\mathrm{P}_F)}$, $\mathrm{AUC}_{(\mathrm{P}_D,\tau)}$, and $\mathrm{AUC}_{(\mathrm{P}_F,\tau)}$, respectively, represent the effectiveness of the detector, the capability of target preservation, and the capability of background suppression. $\mathrm{AUC}_{(\mathrm{ODP})}$ evaluates the overall detection probability, $\mathrm{AUC}_{(\mathrm{SNPR})}$ evaluates the signal-to-noise probability ratio, and $\mathrm{AUC}_{(\mathrm{TDBS})}$ evaluates the combined capabilities of target detection and background suppression. In essence, the lower the value of $\mathrm{AUC}_{(\mathrm{P}_F,\tau)}$, the higher the other values, and the better the detection performance of the HAD detector.

4) Results and Analysis: To qualitatively and quantitatively assess the detection performance, we provide visual observation and several evaluation indicators of the results, i.e., separability boxplot, 2-D and 3-D ROC curves, AUC scores, and time complexity. Below, we will conduct a detailed analysis based on these evaluation criteria one by one.

a) Visual quality: First, the detection maps of different methods on the tested datasets are shown in Fig. 4, form which we observe that our method is capable of finding the locations of the abnormal targets accurately. For example, as the Salinas

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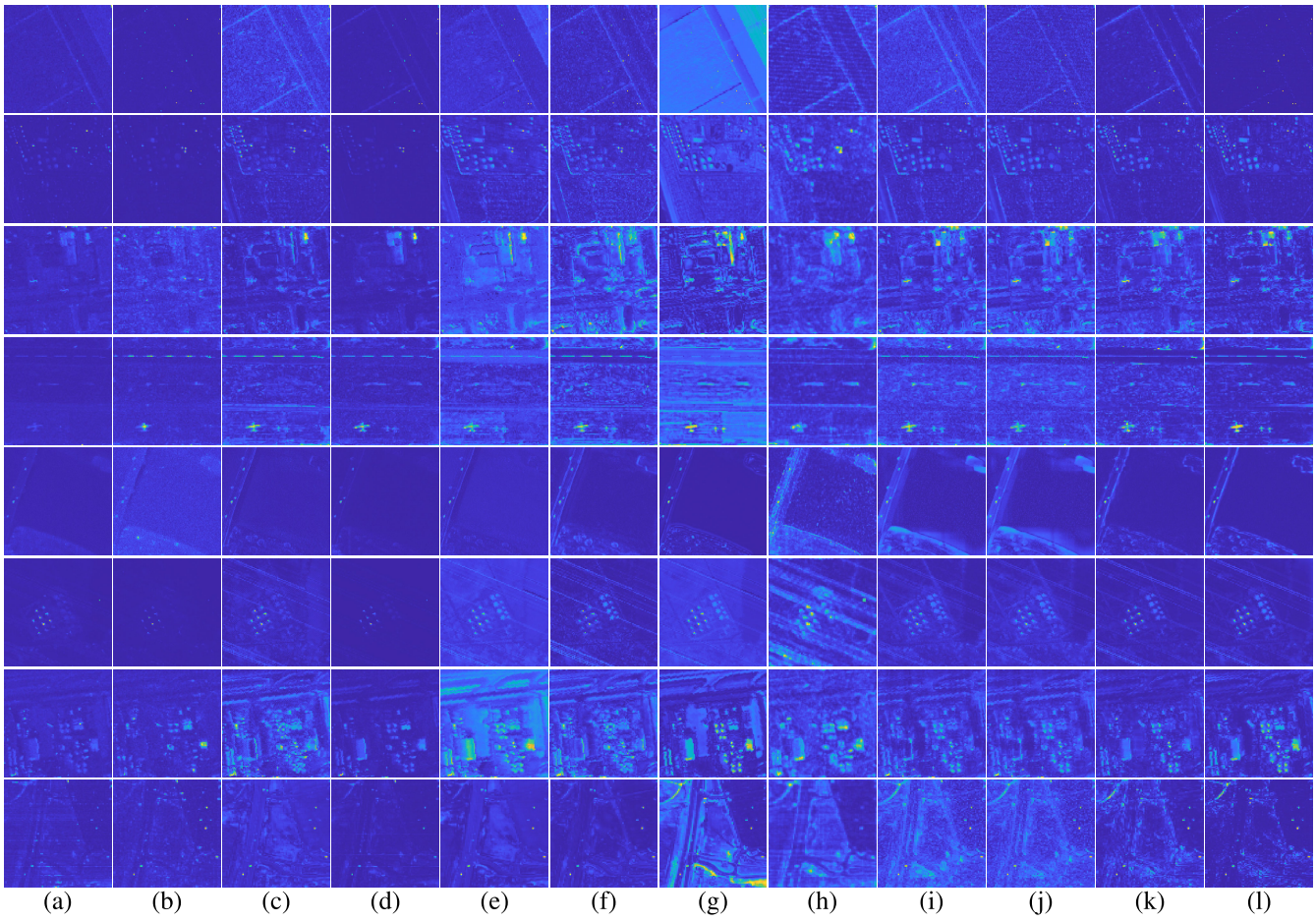


Fig. 4. Anomaly detection map of various HAD methods on eight HSI datasets: Salinas, Segundo, Airport-1, Airport-4, Beach-4, Urban-2, Urban-5, and HYDICE (from top to bottom). (a) RX. (b) CRD. (c) DeCNN-AD. (d) LSMAD. (e) LRASR. (f) GTVLRR. (g) PTA. (h) TPCA. (i) TLRSR. (j) GNLTR. (k) TRDFTVAD. (l) Proposed.

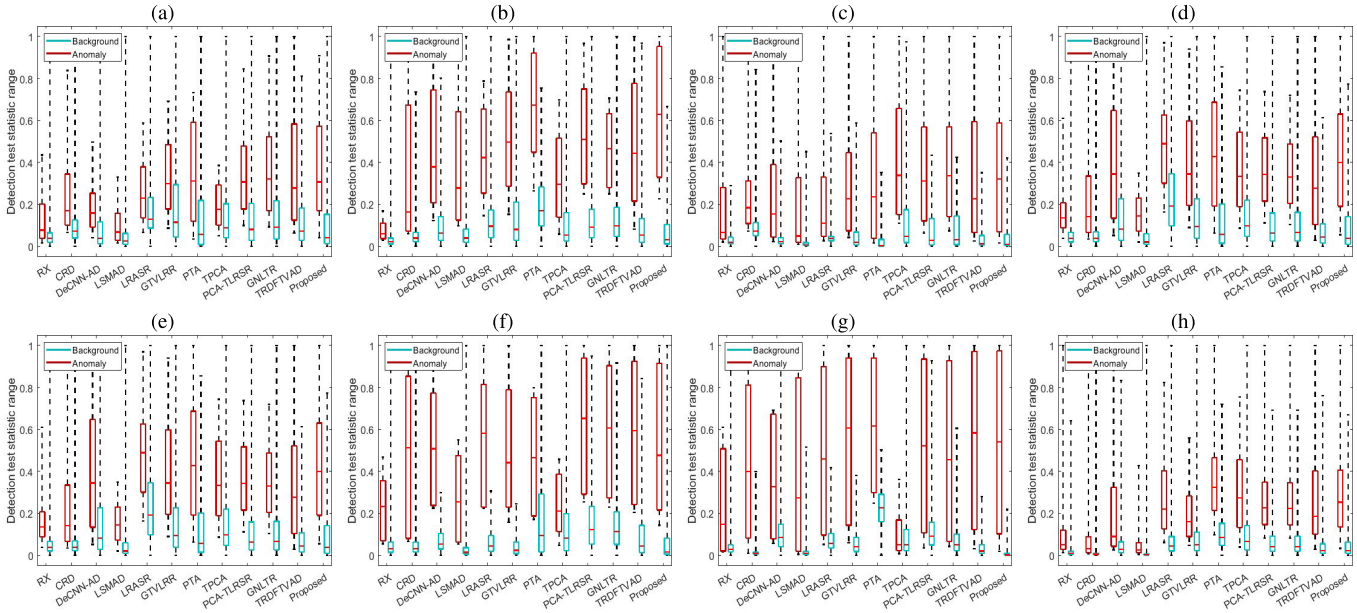


Fig. 5. Separability maps for different HSI datasets. (a) Airport-1. (b) Airport-4. (c) Beach-4. (d) Urban-2. (e) Urban-5. (f) HYDICE. (g) Salinas. (h) Segundo.

dataset shown in the first row of Fig. 4, the RX, DeCNN-AD, PTA, and TPCA fail to detect the anomaly pixels effectively. In comparison to other competitors, the CRD and LSMAD demonstrate superior outcomes in background suppression, yet they unfortunately tend to suppress some abnormal pixels as well. Although the LRASR, GTVLRR, TLRSR, GNLTR, and

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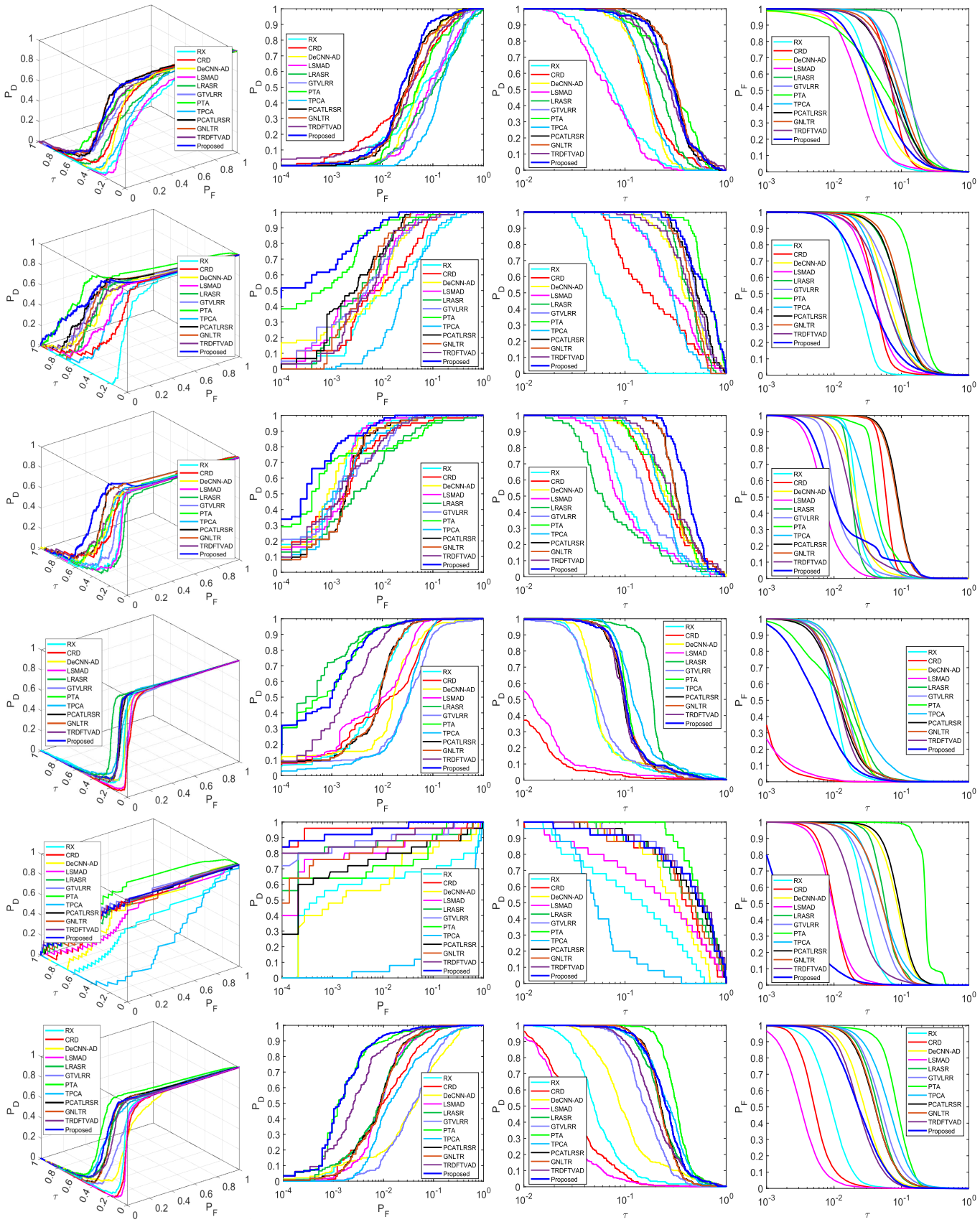


Fig. 6. 3-D and 2-D ROC curves' performance comparison of different HAD approaches on six HSI datasets: Airport-1, Airport-4, Pavia, Urban-4, Salinas, and Segundo (from top to bottom). (a) ROC curves of (P_D, P_F, τ). (b) ROC curves of (P_D, P_F). (c) ROC curves of (P_D, τ). (d) ROC curves of (P_F, τ).

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TABLE I

AUC VALUES AND AVERAGE RUNNING TIME OUTPUT BY VARIOUS HAD METHODS ON 12 HSI DATASETS. THE BEST AND THE SECOND-BEST RESULTS ARE HIGHLIGHTED IN BOLDFACE AND BLUE, RESPECTIVELY

HSI-Name	AUC-Metrics	RX	CRD	DeCNN-AD	LSMAD	LRASR	GTVLRR	PTA	TPCA	TLRSR	GNLTR	TRDFTVAD	Proposed
Airport-1	$AUC_{(P_D,P_F)}$	0.8219	0.9257	0.9158	0.8408	0.8032	0.8656	0.9075	0.8022	0.9455	0.9374	0.9233	**0.9540**
	$AUC_{(P_D,\tau)}$	0.0980	0.2076	0.1708	0.0849	0.2408	0.3108	0.3363	0.1825	0.3346	**0.3468**	0.3181	0.3259
	$AUC_{(P_F,\tau)}$	0.0423	0.0799	0.0567	**0.0361**	0.1495	0.1467	0.0949	0.1127	0.1046	0.1152	0.0958	0.0697
	$AUC_{(ODP)}$	0.8776	1.0534	1.0299	0.8896	0.8944	1.0298	1.1489	0.8720	1.1755	1.1689	1.1455	**1.2102**
	$AUC_{(SNPR)}$	2.3148	2.5984	3.0127	2.3526	1.6107	2.1189	3.5442	1.6191	3.1983	3.0099	3.3181	**4.6774**
	$AUC_{(TDBS)}$	0.0556	0.1277	0.1141	0.0488	0.0913	0.1641	0.2414	0.0698	0.2300	0.2316	0.2222	**0.2562**
Airport-4	$AUC_{(P_D,P_F)}$	0.9525	0.9749	0.9842	0.9884	0.9876	0.9837	0.9940	0.9431	0.9931	0.9925	0.9842	**0.9977**
	$AUC_{(P_D,\tau)}$	0.0727	0.2664	0.4524	0.3499	0.4378	0.5107	**0.6716**	0.2975	0.5411	0.4640	0.4799	0.6285
	$AUC_{(P_F,\tau)}$	**0.0247**	0.0446	0.0793	0.0488	0.1065	0.1045	0.1833	0.0789	0.1048	0.1105	0.0707	0.0499
	$AUC_{(ODP)}$	1.0005	1.1966	1.3572	1.2896	1.3189	1.3899	1.4824	1.1617	1.4294	1.3460	1.3933	**1.5763**
	$AUC_{(SNPR)}$	2.9409	5.9724	5.7031	7.1728	4.1107	4.8888	3.6648	3.7706	5.1652	4.1997	6.7826	**12.5990**
	$AUC_{(TDBS)}$	0.0480	0.2218	0.3731	0.3011	0.3313	0.4062	0.4883	0.2186	0.4364	0.3535	0.4091	**0.5786**
Beach-3	$AUC_{(P_D,P_F)}$	**0.9998**	0.9897	0.9938	0.9992	0.9983	0.9903	0.9993	0.9981	0.9994	0.9985	0.9915	0.9986
	$AUC_{(P_D,\tau)}$	0.4937	0.3588	0.5599	0.5019	0.5604	0.5555	0.5581	0.6044	0.7116	0.6597	0.4982	**0.7132**
	$AUC_{(P_F,\tau)}$	0.0259	0.0550	0.1099	**0.0237**	0.0682	0.1125	0.0293	0.0744	0.0701	0.0804	0.0549	0.0933
	$AUC_{(ODP)}$	1.4675	1.2934	1.4438	1.4774	1.4904	1.4334	1.5281	1.5281	**1.6409**	1.5778	1.4348	1.6184
	$AUC_{(SNPR)}$	19.0481	6.5225	5.0948	**21.2200**	8.2116	4.9395	19.0431	8.1234	10.1481	8.2065	9.0806	7.6402
	$AUC_{(TDBS)}$	0.4678	0.3038	0.4500	0.4782	0.4921	0.4431	0.5288	0.5300	**0.6415**	0.5793	0.4433	0.6198
Beach-4	$AUC_{(P_D,P_F)}$	0.9538	0.9571	0.9661	0.9702	0.9341	0.9760	0.9777	0.9582	0.9643	0.9696	**0.9801**	0.9727
	$AUC_{(P_D,\tau)}$	0.1284	0.2121	0.2071	0.1133	0.1728	0.2526	0.2748	**0.3813**	0.3328	0.3511	0.2860	0.3319
	$AUC_{(P_F,\tau)}$	0.0233	0.0787	0.0259	0.0087	0.0380	0.0302	0.0114	0.0774	0.0540	0.0580	0.0226	0.0228
	$AUC_{(ODP)}$	1.0589	1.0905	1.1474	1.0748	1.0688	1.1984	1.2411	1.2621	1.2431	1.2626	1.2435	**1.2818**
	$AUC_{(SNPR)}$	5.5066	2.6949	8.0043	12.9600	4.5427	8.3670	**24.1260**	4.9265	6.1572	6.0506	12.6500	14.5790
	$AUC_{(TDBS)}$	0.1051	0.1334	0.1813	0.1046	0.1348	0.2225	0.2634	0.3039	0.2787	0.2931	0.2634	**0.3092**
Urban-2	$AUC_{(P_D,P_F)}$	0.9945	0.9610	0.9495	0.9844	0.9937	0.9691	0.9969	0.9408	**0.9982**	0.9980	0.9912	0.9966
	$AUC_{(P_D,\tau)}$	0.1148	0.0429	0.1707	0.0499	0.2705	0.2228	0.2974	**0.3766**	0.2467	0.2523	0.2155	0.2565
	$AUC_{(P_F,\tau)}$	0.0135	**0.0060**	0.0384	0.0029	0.0913	0.0354	0.0873	0.1032	0.0405	0.0453	0.0257	0.0267
	$AUC_{(ODP)}$	1.0958	0.9979	1.0818	1.0314	1.1729	1.1565	1.2070	1.2142	1.2043	1.2050	1.1810	**1.2264**
	$AUC_{(SNPR)}$	8.4948	7.1429	4.4406	**17.1390**	2.9623	6.2971	3.4062	3.6504	6.0834	5.5675	8.3830	9.6054
	$AUC_{(TDBS)}$	0.1012	0.0369	0.1323	0.0470	0.1792	0.1874	0.2101	**0.2734**	0.2061	0.2070	0.1898	0.2298
Urban-4	$AUC_{(P_D,P_F)}$	0.9886	0.9703	0.9663	0.9816	0.9957	0.9427	0.9961	0.9616	0.9882	0.9876	0.9953	**0.9962**
	$AUC_{(P_D,\tau)}$	0.0874	0.0205	0.0858	0.0312	**0.2066**	0.0843	0.1349	0.1634	0.117	0.1187	0.1246	0.1307
	$AUC_{(P_F,\tau)}$	0.0114	**0.0012**	0.0208	0.0012	0.0217	0.0184	0.0156	0.0322	0.0162	0.0171	0.0144	0.0089
	$AUC_{(ODP)}$	1.0647	0.9897	1.0313	1.0117	1.1806	1.0086	1.1153	1.0929	1.0890	1.0892	1.1054	**1.1179**
	$AUC_{(SNPR)}$	7.6677	17.1840	4.1285	**26.1800**	9.5155	4.5845	8.6512	5.0805	7.2406	6.9238	8.6494	14.6310
	$AUC_{(TDBS)}$	0.0760	0.0193	0.0650	0.0301	**0.1849**	0.0659	0.1193	0.1313	0.1008	0.1016	0.1102	0.1217
Urban-5	$AUC_{(P_D,P_F)}$	0.9691	0.9492	0.9100	0.9612	0.9296	0.9412	0.9557	0.9369	**0.9836**	0.9812	0.9615	0.9747
	$AUC_{(P_D,\tau)}$	0.1456	0.1784	0.3667	0.1477	**0.4792**	0.3745	0.4357	0.3524	0.3596	0.3425	0.3063	0.4121
	$AUC_{(P_F,\tau)}$	0.0437	0.0474	0.1091	**0.0319**	0.2166	0.1199	0.0918	0.1219	0.0814	0.0841	0.0579	0.0637
	$AUC_{(ODP)}$	1.0710	1.0802	1.1676	1.0771	1.1923	1.1958	1.2996	1.1673	1.2618	1.2395	1.2099	**1.3231**
	$AUC_{(SNPR)}$	3.3311	3.7634	3.3619	4.6369	2.2130	3.1236	4.7470	2.8911	4.4158	4.0704	5.2928	**6.4708**
	$AUC_{(TDBS)}$	0.1019	0.1310	0.2576	0.1159	0.2627	0.2546	0.3439	0.2305	0.2782	0.2583	0.2485	**0.3484**
Hyperion	$AUC_{(P_D,P_F)}$	0.9978	0.9947	0.9979	**0.9981**	0.9969	0.9934	0.9466	0.8390	0.9935	0.9909	0.9828	0.9831
	$AUC_{(P_D,\tau)}$	0.2184	0.2961	0.3401	0.1691	0.3261	0.3464	**0.7581**	0.3347	0.3783	0.3815	0.4041	0.4541
	$AUC_{(P_F,\tau)}$	0.0330	0.0561	0.0434	**0.0093**	0.0481	0.0550	0.4996	0.1538	0.0555	0.0701	0.0739	0.0865
	$AUC_{(ODP)}$	1.1832	1.2347	1.2946	1.1579	1.2749	1.2847	1.2051	1.0198	1.3162	1.3024	1.3130	**1.3508**
	$AUC_{(SNPR)}$	6.6090	5.2806	7.8434	**18.1100**	6.7861	6.2994	1.5174	2.1758	6.8134	5.4464	5.4695	5.2516
	$AUC_{(TDBS)}$	0.1854	0.2400	0.2967	0.1598	0.2780	0.2914	0.2585	0.1809	0.3228	0.3115	0.3302	**0.3677**
HYDICE	$AUC_{(P_D,P_F)}$	0.9762	0.9888	**0.9997**	0.9923	0.9995	**0.9997**	0.9131	0.8621	0.9892	0.9912	0.9920	0.9925
	$AUC_{(P_D,\tau)}$	0.2170	0.4550	0.5298	0.2458	0.5435	0.4972	0.4769	0.2420	**0.6214**	0.5953	0.5594	0.5442
	$AUC_{(P_F,\tau)}$	0.0381	0.0378	0.0610	**0.0195**	0.0520	0.0315	0.1379	0.1014	0.1367	0.1246	0.0667	0.0381
	$AUC_{(ODP)}$	1.1552	1.4060	1.4685	1.2186	1.4910	1.4654	1.2520	1.0025	1.4740	1.4619	1.4847	**1.4985**
	$AUC_{(SNPR)}$	5.7134	12.0430	8.6827	12.6010	10.4450	**15.7960**	3.4573	2.3852	4.5475	4.7768	8.3858	14.2643
	$AUC_{(TDBS)}$	0.1790	0.4172	0.4688	0.2263	0.4915	0.4657	0.3389	0.1405	0.4848	0.4707	0.4927	**0.5060**
Pavia-Center	$AUC_{(P_D,P_F)}$	0.9981	0.9945	0.9961	0.9979	0.9779	0.9946	0.9879	0.9957	0.9970	0.9970	0.9950	**0.9994**
	$AUC_{(P_D,\tau)}$	0.1884	0.3518	0.3105	0.1859	0.1522	0.2466	0.3777	0.3100	0.4268	0.4242	0.4011	**0.5134**
	$AUC_{(P_F,\tau)}$	0.0233	0.0618	0.0256	**0.0091**	0.0199	0.0158	0.0525	0.0396	0.0942	0.0914	0.0276	0.0324
	$AUC_{(ODP)}$	1.1632	1.2845	1.2809	1.1747	1.1101	1.2254	1.3131	1.2661	1.3295	1.3298	1.3684	**1.4803**
	$AUC_{(SNPR)}$	8.0852	5.6936	12.1120	**20.4310**	7.6289	15.5690	7.1961	7.8308	4.5302	4.6406	14.5370	15.8270
	$AUC_{(TDBS)}$	0.1651	0.2900	0.2849	0.1768	0.1322	0.2308	0.3252	0.2704	0.3326	0.3328	0.3735	**0.4810**
Salinas	$AUC_{(P_D,P_F)}$	0.8074	0.9809	0.9030	0.9522	0.9483	0.9804	0.9707	0.5181	0.9226	0.9460	0.9871	**0.9984**
	$AUC_{(P_D,\tau)}$	0.2045	0.4386	0.3486	0.3543	0.4834	0.5589	**0.6143**	0.0776	0.5299	0.4809	0.5640	0.5444
	$AUC_{(P_F,\tau)}$	0.0313	0.0106	0.0928	0.0114	0.0632	0.0475	0.2346	0.0654	0.0997	0.0578	0.0280	**0.0043**
	$AUC_{(ODP)}$	0.9805	1.4089	1.1589	1.2950	1.3685	1.4918	1.3504	0.5303	1.3528	1.3691	1.5230	**1.5385**
	$AUC_{(SNPR)}$	6.5244	41.3440	3.7576	30.9860	7.6472	11.7690	2.6190	1.1860	5.3149	8.3232	20.1430	**126.4500**
	$AUC_{(TDBS)}$	0.1731	0.4280	0.2559	0.3428	0.4202	0.5114	0.3798	0.0122	0.4302	0.4231	0.5360	**0.5401**
Segundo	$AUC_{(P_D,P_F)}$	0.9828	0.9744	0.9158	0.9824	0.9816	0.9310	0.9929	0.9512	0.9859	0.9856	0.9909	**0.9941**
	$AUC_{(P_D,\tau)}$	0.0725	0.0440	0.1433	0.0334	0.2510	0.1795	**0.3408**	0.2828	0.2434	0.2405	0.2314	0.2760
	$AUC_{(P_F,\tau)}$	0.0113	0.0069	0.0360	**0.0046**	0.0531	0.0620	0.0937	0.0791	0.0506	0.0509	0.0297	0.0312
	$AUC_{(ODP)}$	1.0441	1.0116	1.0231	1.0112	1.1794	1.0485	**1.2400**	1.1550	1.1786	1.1752	1.1926	1.2389
	$AUC_{(SNPR)}$	6.3920	6.4150	3.9798	7.2213	4.7235	2.8958	3.6381	3.5777	4.8088	4.7241	7.7953	**8.8551**
	$AUC_{(TDBS)}$	0.0612	0.0371	0.1073	0.0288	0.1978	0.1175	**0.2471**	0.2038	0.1928	0.1896	0.2017	0.2448
Average Time (second)		**0.2858**	7.8112	94.5535	18.3238	65.7009	195.1293	76.1356	41.8671	11.1041	12.0203	117.3451	94.4689

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TABLE II

AUC$_{(ODP)}$ VALUES OBTAINED BY VARIOUS NONCONVEX COMBINATIONS $\Phi(\cdot)+\psi(\cdot)$ FOR DIFFERENT HSI DATASETS

HSI-Name	$h(\cdot)$	Firm+Firm	MCP+MCP	SCAD+SCAD	ℓ_q+ℓ_q	ℓ_q+Firm	ℓ_q+MCP	ℓ_q+SCAD	Log+ℓ_q	Log+Firm	Log+MCP	Log+SCAD
Airport-4	ℓ_1-norm	1.5481	1.5561	1.5519	1.5407	1.5444	1.5444	1.4450	1.5682	1.5677	1.5677	1.3060
	$\ell_{F,1}$-norm	1.5708	1.5763	1.5778	1.5527	1.5631	1.5631	1.4645	1.5723	1.5719	1.5719	1.2399
Beach-4	ℓ_1-norm	1.2659	1.2644	1.2674	1.2751	1.2753	1.2753	1.2749	1.2745	1.2748	1.2748	1.2804
	$\ell_{F,1}$-norm	1.2804	1.2818	1.2818	1.2782	1.2877	1.2877	1.2717	1.2786	1.2799	1.2799	1.2855
Urban-2	ℓ_1-norm	1.2136	1.2147	1.2163	1.2107	1.2111	1.2111	1.2152	1.2286	1.2279	1.2279	1.1991
	$\ell_{F,1}$-norm	1.2245	1.2264	1.2238	1.2167	1.2182	1.2182	1.2128	1.2289	1.2289	1.2289	1.1513
Urban-4	ℓ_1-norm	1.1153	1.1158	1.1146	1.1152	1.1155	1.1155	1.1133	1.1139	1.1150	1.1150	1.1666
	$\ell_{F,1}$-norm	1.1185	1.1194	1.1160	1.1159	1.1182	1.1182	1.1128	1.1146	1.1147	1.1147	1.1599
Hyperion	ℓ_1-norm	1.3277	1.3306	1.3354	1.3368	1.3383	1.3383	1.3389	1.3556	1.3519	1.3519	1.2629
	$\ell_{F,1}$-norm	1.3378	1.3386	1.3415	1.3415	1.3443	1.3443	1.3339	1.3517	1.3508	1.3508	1.2491
HYDICE	ℓ_1-norm	1.4664	1.4651	1.4344	1.4889	1.4847	1.4847	1.4687	1.4618	1.4488	1.4488	1.5152
	$\ell_{F,1}$-norm	1.4749	1.4731	1.4348	1.4885	1.4996	1.4996	1.4696	1.4540	1.4487	1.4487	1.4722
Pavia-Center	ℓ_1-norm	1.4743	1.4791	1.4711	1.4726	1.4746	1.4746	1.4514	1.4667	1.4645	1.4645	1.4499
	$\ell_{F,1}$-norm	1.4773	1.4803	1.4728	1.4778	1.4770	1.4770	1.4467	1.4669	1.4666	1.4666	1.4580
Salinas	ℓ_1-norm	1.5163	1.5166	1.5149	1.5174	1.5170	1.5170	1.5140	1.5162	1.5193	1.5193	1.3076
	$\ell_{F,1}$-norm	1.5366	1.5395	1.5283	1.5255	1.5304	1.5304	1.5363	1.5200	1.5197	1.5197	1.3083

TABLE III

AUC$_{(ODP)}$ VALUES OBTAINED BY VARIOUS NONCONVEX COMBINATIONS $\Phi(\cdot)+\psi(\cdot)$ FOR DIFFERENT NOISY HSI DATASETS

HSI-Name	σ	Firm+Firm	MCP+MCP	SCAD+SCAD	ℓ_q+ℓ_q	ℓ_q+Firm	ℓ_q+MCP	ℓ_q+SCAD	Log+ℓ_q	Log+Firm	Log+MCP	Log+SCAD
Airport-4	0.001	1.5760	1.5779	1.5777	1.5477	1.5671	1.5706	1.5411	1.5730	1.5716	1.5730	1.1992
	0.005	1.5569	1.5575	1.5638	1.5387	1.5521	1.5535	1.5301	1.5580	1.5534	1.5533	1.1063
	0.01	1.5441	1.5448	1.5503	1.5264	1.5378	1.5408	1.5251	1.5458	1.5446	1.5396	1.0352
Beach-4	0.001	1.2661	1.2662	1.2638	1.2615	1.2771	1.2756	1.2702	1.2657	1.2679	1.2675	1.2786
	0.005	1.2532	1.2554	1.2582	1.2518	1.2646	1.2641	1.2627	1.2583	1.2559	1.2587	1.2746
	0.01	1.2501	1.2484	1.2501	1.2477	1.2537	1.2530	1.2593	1.2512	1.2592	1.2506	1.2609
Urban-2	0.001	1.2197	1.2192	1.2153	1.2055	1.2148	1.2147	1.2095	1.2220	1.2213	1.2222	1.2099
	0.005	1.2079	1.2072	1.2043	1.2003	1.2024	1.2053	1.1980	1.2101	1.2104	1.2113	1.2519
	0.01	1.2004	1.2007	1.1979	1.1944	1.1966	1.1977	1.1930	1.2033	1.2027	1.2045	1.3055
Urban-4	0.001	1.1120	1.1067	1.1070	1.1062	1.1124	1.1129	1.1150	1.1044	1.1097	1.1095	1.1865
	0.005	1.0966	1.0959	1.0955	1.0953	1.1016	1.0998	1.0931	1.0949	1.1026	1.1003	1.1846
	0.01	1.0926	1.0878	1.0892	1.0886	1.0953	1.0964	1.0857	1.0870	1.1118	1.1036	1.1977
Hyperion	0.001	1.3265	1.3241	1.3313	1.3211	1.3409	1.3434	1.3377	1.3467	1.3491	1.3518	1.2995
	0.005	1.3087	1.3046	1.3116	1.3214	1.3245	1.3293	1.3240	1.3325	1.3379	1.3340	1.2636
	0.01	1.3001	1.2877	1.2956	1.3104	1.3119	1.3165	1.3137	1.3193	1.3182	1.3171	1.2198
HYDICE	0.001	1.3953	1.4130	1.4367	1.4099	1.3825	1.4127	1.4510	1.4226	1.4234	1.4299	1.4766
	0.005	1.3976	1.3958	1.4012	1.4318	1.4247	1.3935	1.4232	1.3894	1.3878	1.3877	1.4363
	0.01	1.3796	1.3812	1.3828	1.4174	1.4001	1.3875	1.4089	1.3712	1.3612	1.3609	1.3495
Pavia-Center	0.001	1.4771	1.4769	1.4738	1.4711	1.4767	1.4770	1.4537	1.4602	1.4606	1.4604	1.4541
	0.005	1.4673	1.4672	1.4624	1.4675	1.4643	1.4662	1.4431	1.4599	1.4550	1.4571	1.4559
	0.01	1.4620	1.4631	1.4601	1.4622	1.4593	1.4604	1.4386	1.4585	1.4558	1.4579	1.4658
Salinas	0.001	1.5163	1.5157	1.5162	1.5086	1.5100	1.5067	1.5128	1.5162	1.5225	1.5188	1.3478
	0.005	1.5017	1.5012	1.5034	1.5005	1.5023	1.5021	1.5013	1.4915	1.4992	1.5023	1.3508
	0.01	1.4878	1.4872	1.4895	1.4863	1.4865	1.4861	1.4921	1.4884	1.4866	1.4880	1.3665

TRDFTVAD are capable of detecting a significant portion of anomaly pixels, they are susceptible to some degree of background interference. The proposed method strikes a superior balance between accurate anomaly recognition and effective background removal over other detectors. For the Airport-4 dataset presented in the fourth row of Fig. 4, the RX performs the worst in the identification of three anomalous aircrafts. The CRD and TPCA cannot successfully detect two small anomalous planes. On the contrary, the DeCNN-AD, LSMAD, LRASR, and TRDFTVAD can detect these two small planes, but their overall pictures are relatively poor. For the detection of all abnormal aircrafts, the GTVLRR, PTA, TLRSR, GNLTR, and our detector can yield better results compared to others. Importantly, our approach possesses a strong ability in highlighting the targeted anomalies and suppressing the background. For the Segundo dataset displayed in the second row of Fig. 4, the RX, CRD, DeCNN-AD, and LSMAD have low responses to the abnormal pixels. The GTVLRR and TPCA can identify most of the abnormal pixels but with indistinct contours. The PTA performs the best in detecting abnormal objects, but it is susceptible to clearbackground. The LRASR, TLRSR, GNLTR, TRDFTVAD, and our detector are able to precisely locate clear anomalous objects while suppressing the background, among which the proposed one exhibits the utmost proficiency.

b) Separability boxplot: Second, the normalized background-anomaly separation maps are employed to showcase the capabilities of various methods in separating abnormal pixels from background pixels. The separation maps reflect the statistical characteristics of anomaly-background pixels within the detection outcomes. As shown in Fig. 5, the boxes represent the pixel ranges of 10%–90%, and the horizontal line in each box signifies the median value of the pixels. The lines above and below the boxes represent the 90%–100% and 0%–10% intervals, respectively. The wider the gaps between the anomaly and background boxes, the greater the distinguishability achieved by the HAD detector. As depicted in Fig. 5, the background-anomaly separation of the proposed method is better than that of the competitive algorithms in the majority of cases. In rare cases, our method has the similar separable capability as some competing methods, i.e., DeCNN-AD, LRASR, GTVLRR, PTA, PCA-TLRSR, GNLTR, and TRDFTVAD. Specifically, as presented in Fig. 5(a)–(c), (g), and (h), our method

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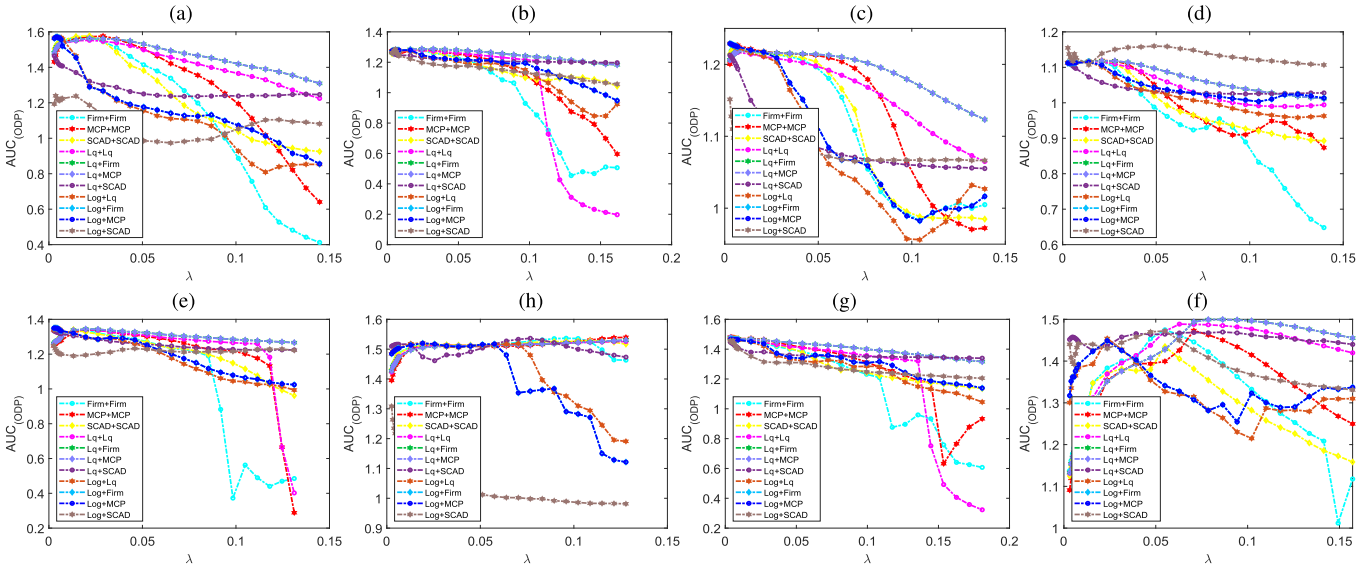


Fig. 7. Influence of various regularization parameters λ upon HAD performance under different nonconvex functions $\Phi(\cdot)$ and $\psi(\cdot)$. For the anomaly-sparsity regularization item, $h(\cdot) = \|\cdot\|_{F,1}$ is defined as the $\ell_{F,1}$-norm. (a) Airport-4. (b) Beach-4. (c) Urban-2. (d) Urban-4. (e) Hyperion. (f) HYDICE. (g) Pavia-Center. (h) Salinas.

has the largest distance between the blue and red boxes. Besides, the median of the background is very close to zero, which demonstrates that our method owns a strong ability to suppress backgrounds. The PTA, PCA-TLRSR, GNLTR, TRDFTVAD, and our algorithm obtain the comparable separation gap in Fig. 5(h). Furthermore, similar separation intervals are achieved by the PCA-TLRSR, GNLTR, and our algorithm in Fig. 5(d) and (e). As shown in Fig. 5(f), the DeCNN-AD, LRASR, and GTVLRR algorithms achieve better separation than our approach. However, they perform poorly on other HSI datasets. Overall, the proposed HAD algorithm achieves a good separation and there is no overlap between the anomaly-background boxes on the tested datasets.

c) ROC curves: Third, the 3-D and 2-D ROC curves of twelve HAD algorithms on all tested datasets are presented in Fig. 6. From the ROC curves plotted in Fig. 6(b), we observe that among the vast majority of P_F values (i.e., false alarm probability), the proposed method can achieve the highest P_D values (i.e., detection probability) on most HSI datasets. The 2-D ROC curve of $(\mathrm{P}_D, \mathrm{P}_F)$ demonstrates better overall detection performance when it is positioned closer to the top-left corner. As presented in Fig. 6(c), it can be found that compared with some detectors (e.g., RX, CRD, DeCNN-AD, LSMAD, LRASR, GTVLRR, and TRDFTVAD), the 2-D ROC curves of (P_D, τ) induced by our detector are closer to the top-right corner in most cases, which reflects its stronger target detection ability. With regard to the 2-D ROC curve of (P_F, τ), the closer the curve is to the bottom-left corner, the better the background suppression ability. As provided in Fig. 6(d), compared with tensor-based detectors, the proposed one achieves the optimal background suppression effect on almost all datasets. On the whole, the background suppression effect of the proposed algorithm on all datasets is at the upper middle level over other HAD algorithms.

d) AUC scores, time complexity: Different from the ROC curves, which presents the relationship among P_D, P_F, and τ, the AUC values can directly reflect the performance of different methods. Based on the 3-D ROC curves, various AUC scores of all anomaly detectors on twelve HSI datasets are provided in Table I, where the best AUC values are highlighted in boldface while the second highest AUC values are highlighted in blue underline. Strikingly, as shown in Table I, the proposed method reaches the highest $\mathrm{AUC}_{(\mathrm{ODP})}$ values (an overall detection performance metric) in most cases except the Beach-3 and Segundo. The values of $\mathrm{AUC}_{(\mathrm{P}_D,\mathrm{P}_F)}$ also achieve the highest in most datasets. Overall, most of the AUC scores achieved by our method on each HSI dataset can rank first or second, which indicates that our HAD algorithm possesses the best comprehensive performance. In addition, Table I also gives the average running time of different algorithms on all tested datasets. It can be seen that the RX has the lowest computational cost, followed by the CRD, TLRSR, and GNLTR detectors. Note that due to the inclusion of PCA preprocessing scheme, the TLRSR and GNLTR exhibit the shortest running time among the tensor-based approaches.

C. Related Discussions

Parameters Setting: In Section III-C, we list some popular nonconvex penalty functions satisfying the assumptions listed in Definition 6, and their corresponding proximity operators. In our discussions, the parameters of the nonconvex functions should be set in advance. For the firm thresholding, SCAD penalty, and MCP penalty, we respectively set $a, b, c \in \{0.4, 0.6, 0.8, 1, 1.4, 2\}$, $\gamma = \xi = \eta = 1/\mu^{\{\nu+1\}}$ in $\Phi(\cdot)$ and $\psi(\cdot)$ for simplicity. For the ℓ_q penalty, we just need $0 < q < 1$ and simply set $q = 0.9$ in our experiments. For the Log penalty, $\theta > 0$ is just required and θ is stable for the detection performance. We simply set $\theta \in \{0.1, 0.01, 0.001\}$ in this experiment. In addition, for the important trade-off parameter $\lambda > 0$, we empirically set $\lambda = \kappa/(\max(n_1, n_2) \cdot n_3)^{1/2}$, in which $\kappa \in \{0.04{:}0.01{:}0.09, 0.1{:}0.1{:}2\}$, if $h(\cdot) = \|\cdot\|_1$; $\kappa \in \{0.4{:}0.1, 0.9, 1{:}1{:}20\}$, if $h(\cdot) = \|\cdot\|_{F,1}$.

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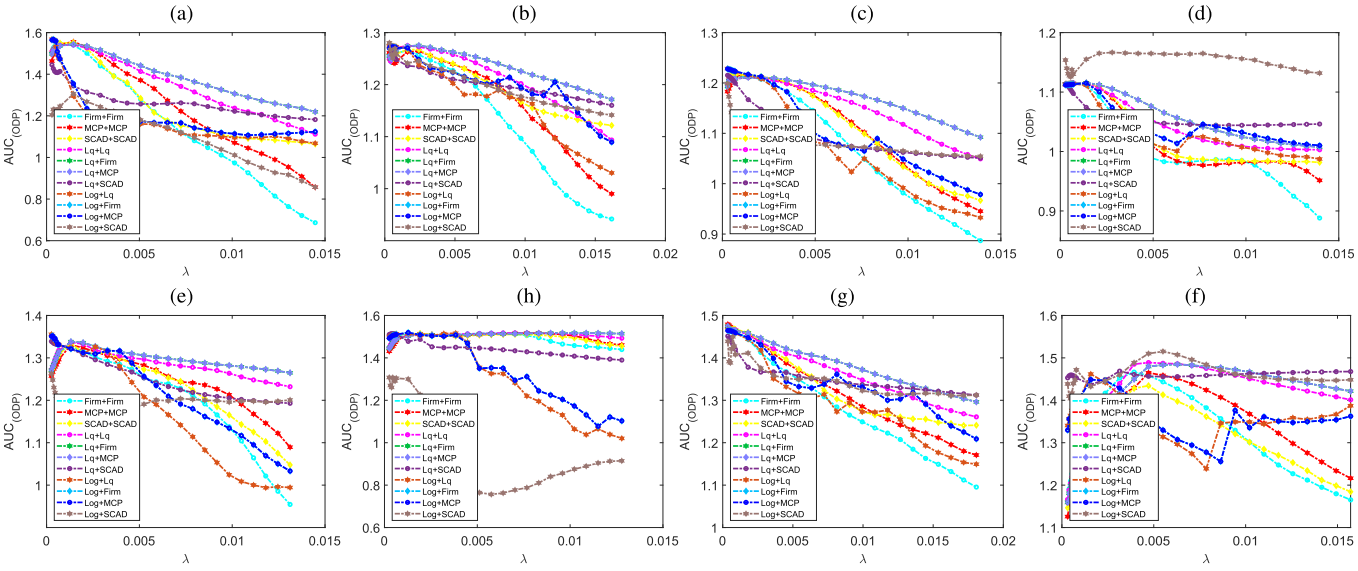


Fig. 8. Influence of various regularization parameters λ upon HAD performance under different nonconvex functions $\Phi(\cdot)$ and $\psi(\cdot)$. For the anomaly-sparsity regularization item, $h(\cdot) = \|\cdot\|_1$ is defined as the ℓ_1-norm. (a) Airport-4. (b) Beach-4. (c) Urban-2. (d) Urban-4. (e) Hyperion. (f) HYDICE. (g) Pavia-Center. (h) Salinas.

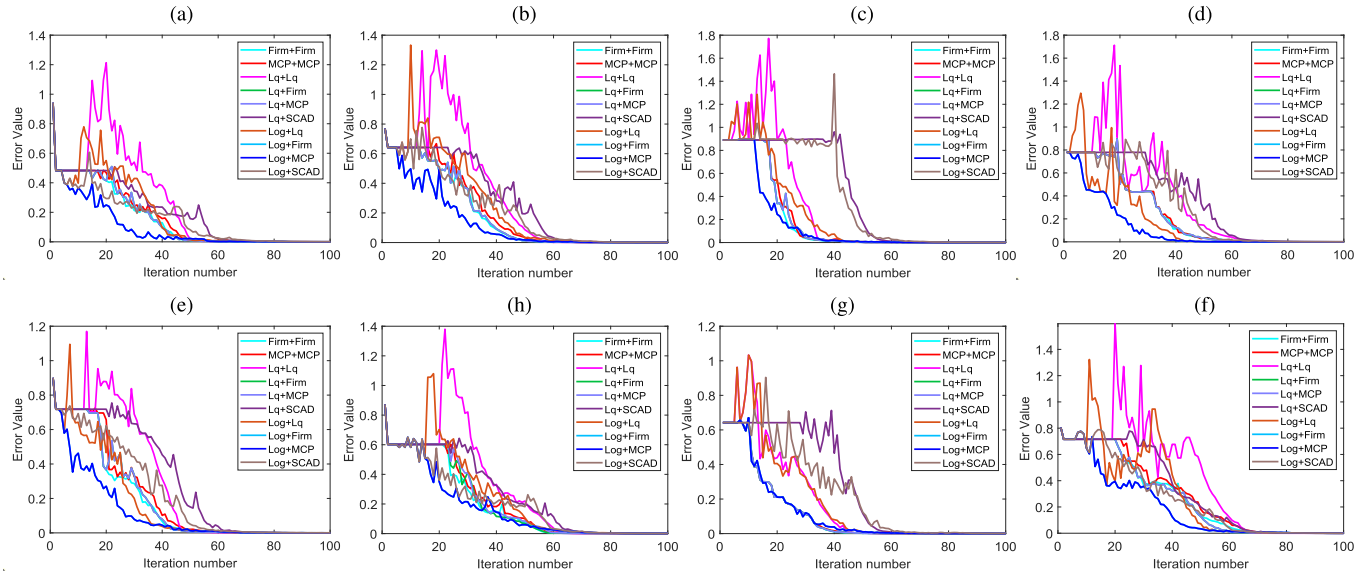


Fig. 9. Convergence behavior of the proposed HAD algorithm under different nonconvex functions $\Phi(\cdot)$ and $\psi(\cdot)$. (a) Airport-4. (b) Beach-4. (c) Urban-2. (d) Urban-4. (e) Hyperion. (f) Salinas. (g) Pavia-Center. (h) HYDICE.

1) Discussion on Different Regularization Parameters λ: In this experiment, we mainly investigate the influence of various regularization parameters λ upon HAD performance under different combinations of nonconvex functions, i.e., $\Phi(\cdot) + \psi(\cdot)$, and anomaly-sparsity regularization items, i.e., $h(\cdot)$. Note that $\Phi(\cdot)$ and $\psi(\cdot)$ could be the same. The nonconvex penalty functions in our experiments include firm, SCAD, MCP, Log, and ℓ_q, where $\Phi(\cdot)$ and $\psi(\cdot)$ are the same nonconvex functions in each case. For different $\Phi(\cdot)$ and $\psi(\cdot)$, we test seven cases: Log + firm, Log + MCP, Log + ℓ_q, Log + SCAD, ℓ_q + firm, ℓ_q + MCP, and ℓ_q + SCAD. Besides, $h(\cdot)$ is divided into two cases, i.e., $\ell_{F,1}$-norm and ℓ_1-norm. The HAD performance is measured by an overall evaluation metric, i.e., $\text{AUC}_{(\text{ODP})}$. The relevant experimental results are displayed in Figs. 7 and 8, from which we can find that for various nonconvex combinations, the proposed algorithm is insensitive to the trade-off parameter λ over a relatively large range on most HSI datasets. In addition, other important observations are that **1)** compared with other nonconvex combinations, Log + SCAD exhibit extremely outstanding performance in rare cases; **2)** in most situations, ℓ_q + ℓ_q, MCP + MCP, ℓ_q + SCAD, ℓ_q + MCP, and ℓ_q + firm show similar HAD performance. Note that the curves of ℓ_q + MCP and ℓ_q + firm have been overlapped.

2) Discussion on Different Nonconvex Penalty Functions $\Phi(\cdot)$ and $\psi(\cdot)$: In this testing case, we mainly verify the performance under different choices of nonconvex functions $\Phi(\cdot)$ and $\psi(\cdot)$ in our HAD model. According to the optimal regularization parameters λ derived from Figs. 7 and 8, Table II reports the $\text{AUC}_{(\text{ODP})}$ values corresponding to various nonconvex combinations on eight HSI datasets. From these

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experimental results, we can see that under the two different sparsity measures, there is no great difference in the $\mathrm{AUC}_{(\mathrm{ODP})}$ values obtained by different nonconvex combinations on the Airport-4, Urban-2, Hyperion, Pavia, and Salinas datasets except for Log + SCAD. Only on the Beach-4 and HYDICE datasets, Log + SCAD exhibit a good HAD performance over other combinations. These findings suggest that our generalized HAD approach is well-structured and performs excellently, allowing for the selection of nonconvex penalty functions that are suited to the HSI datasets.

3) Discussion on Different Sparsity Measures $\psi(h(\cdot))$: In this discussion, we mainly study the HAD performance of two anomaly-sparsity regularizers $\psi(\|\cdot\|_1)$ and $\psi(\|\cdot\|_{F,1})$. From the results shown in Table II, we can conclude that the latter's performance is superior to the former in most cases. Therefore, in our experiments, we tend to favor the latter for capturing the sparsity of anomaly tensors.

4) Discussion on Robustness in Noisy Scenarios: Since HSI data is inevitably affected by noise during data acquisition and transmission, it becomes more challenging to accurately identify anomalous targets on the ground. To address this issue, we formulate novel HAD model that operates effectively in the presence of noise corruptions, i.e.,

$$\begin{aligned}&\min_{\mathcal{L},\mathcal{E}} \ \frac{1}{3}\sum_{t=1}^{3}\left\{\sum_{k=1}^{2} w_k \cdot \Phi\left(\mathcal{G}_t^{\langle k,l\rangle}\right)\right\} + \lambda \cdot \psi\left(h(\mathcal{E})\right) + \beta \cdot \|\mathcal{N}\|_F^2 \\ &\text{s.t. } \mathcal{G}_t = \nabla_t(\mathcal{L}), \quad \mathcal{L}+\mathcal{E}+\mathcal{N} = \mathcal{M} \end{aligned} \tag{22}$$

where the last term denotes the noise regularization. The derivation of ADMM algorithm corresponding to the HAD model (22) is similar to that of model (7).

In order to verify the detection performance and robustness of the proposed method, we conduct related experiments on HSI datasets corrupted by different noise levels. In our experiments, we construct three sets of HSI datasets with different noise intensities by adding noise to each band of each HSI. The added noise is Gaussian noise with zero mean and variances of 0.001, 0.005, and 0.01, respectively. Regarding the model parameters, we set $h(\cdot) = \|\cdot\|_{F,1}$, $\beta \in \{0.001, 0.003, 0.005, 0.01, 0.03, 0.05, 0.1\}$, and the value of λ is the same as that in the previous experiment. From the results presented in Table III, we can find that the proposed HAD algorithm is robust against noisy backgrounds.

5) Discussion on Convergence Behavior: Under different nonconvex combinations, $\Phi(\cdot) + \psi(\cdot)$, the convergence error curves of our HAD algorithm on all eight tested HSI datasets are presented in Fig. 9. It can be seen that although there exist some fluctuations during the first 40 iterations, they finally converge to zero after at most 60 iterations. Thereby, the proposed HAD algorithm has good convergence and stability.

V. Conclusion and Future Work

In this article, based on efficient tensor ring framework, a novel generalized nonconvex approach is developed for hyperspectral anomaly detection. To provide an improved estimate of the HSI's background component, we delve into a novel prior characterization strategy. On this basis, we put forward a concise and unified nonconvex regularizer to characterize the global low-rankness and local smoothness priors of HSI's background simultaneously. Unlike the existing studies that handle these priors separately, our approach avoids the complexity of trade-off parameter adjustment. Additionally, we investigate a new sparsity measure scheme for the HSI's anomaly component, which is encoded by another unified nonconvex regularizer. Extensive experimental results on twelve HSI datasets demonstrate that our method named TR-GNHAD achieves an overall superior anomaly detection performance by effectively suppressing the background distribution. Future work will focus on developing fast and accurate anomaly detection methods for large-scale HSI datasets, leveraging randomized sketching technology [88], [89] to accelerate computation and optimization involving alternating minimization or SVD.

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